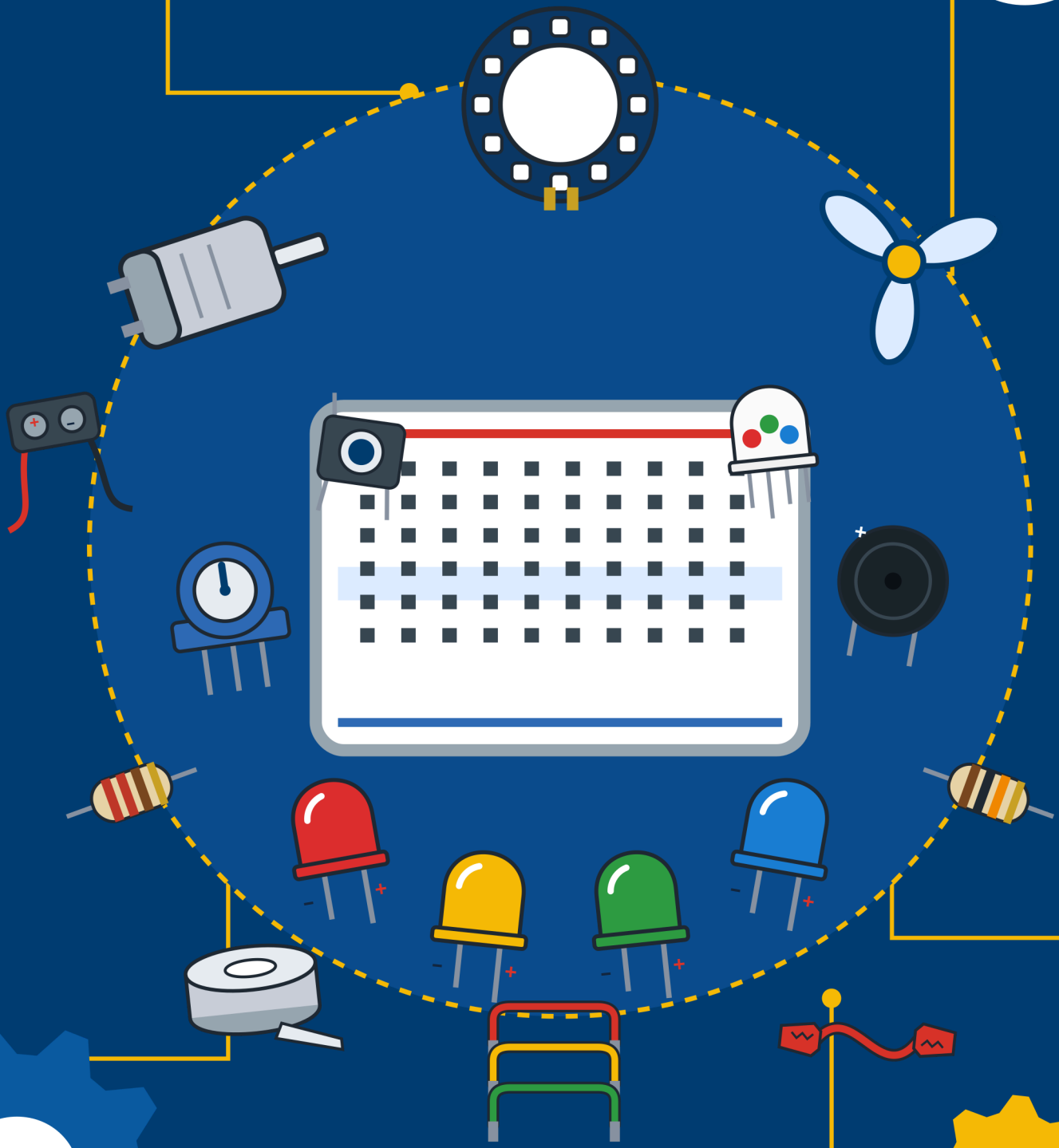
Electrical Circuits Kit

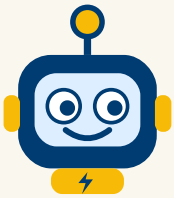
Manual



BUILD. EXPERIMENT. INVENT.

Your first step into the world of electronics.

WELCOME TO YOUR INVENTOR'S JOURNEY



You are about to become an engineer, an inventor and a problem-solver.

Inside this box is everything you need to build real electronic circuits. You will light up LEDs, make a fan spin, send secret signals and build alarms.

You will not just copy circuits. You will predict what will happen, test it, make mistakes and fix them. That is exactly how real engineers work.

By Mission 20 you will design your own invention from scratch.

HOW EACH MISSION WORKS

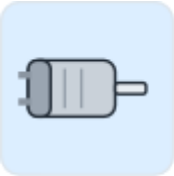
1 · READ	2 · BUILD	3 · DISCOVER	4 · INVENT
the mission brief.	follow the numbered steps.	find out what is happening.	change it and make it yours.

FOUR STAGES TO MASTER

CIRCUIT EXPLORER	INTERACTIVE ENGINEER	MOTION MAKER	MASTER INVENTOR
Missions1–5.Lightsand circuits.	Missions 6–10. Logic and signals.	Missions11–15.Motors and machines.	Missions 16–20. Your own ideas.

SAFETY FIRST. Younger inventors should build with a grown-up nearby. Always connect the battery last and disconnect it when you finish. Never let the red and black battery wires touch each other. Never put any part of the kit into a wall socket. Keep small parts away from babies and pets. Do not squeeze or open batteries.

MEET YOUR COMPONENTS



×1

DC MOTOR

What it does: Turns electricity into spinning.
How it works: Electricity makes magnets inside push the shaft round. Swap the two wires and it spins the other way.

Handle withcare: Do not hold the shaft while it runs. Attach crocodile wires to the two metal tabs.



×1

MINI FAN PROPELLER

What it does: Pushes airto makea breeze.

How it works: Slanted blades scoop air like a paddle scoops water when the motor turns them.

Handle with care: Press it gently onto the motor shaft. Keep fingers clear when it spins.



×1

9V CONNECTOR

What it does: Links a9V battery to your circuit.

How it works: Red wire = positive (+). Black wire = negative (–). The battery pushes electricity out of +.

Handlewithcare: Connectit LAST.Never touch red and black together. The battery is not in the kit.



×1

BREADBOARD

What it does: Aboard with holes that connect parts without soldering.

How it works: Metal strips under the holes join them. See the Breadboard Guide for the full map.

Handle with care: Pushlegsingentlyand straight. Do not force thick wires.



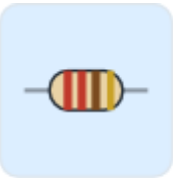
1 strip

JUMPER WIRES

What it does: Move electricity from one hole to another.

How it works: Bare metal ends push into holes. The coloured plastic stops wires touching.

Handle with care: Hold by the plastic end, not the wire. Use red for + and dark for –.



10 total

RESISTORS 220Ω & 10KΩ

What it does: Slow the flow of electricity.
How it works: Ω (ohm) measures how hard a part makes it for electricity to flow. A bigger number = stronger slowing.

Handle with care: Colour bands showthe value (see Resistor Guide). Legs bend; resistors have no + or –.



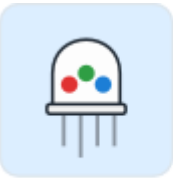
8 total

LEDS: RED, GREEN, BLUE, YELLOW

What it does: Light-emitting diodes: tiny lamps.

How it works: Electricity flows through only one way and makes light. They need a resistor to stay safe.

Handle withcare: Longleg=+ (anode).Short leg and flat edge = – (cathode). Never connect straight to a battery.



×1

RGB LED

What it does: Three LEDs in one: red, green, blue.

How it works: Each colour has its own leg and they share one common leg. Mix the lights to make new colours.

Handle with care: The longest leg isthe shared one. It needs a resistor on each colour.

MEET YOUR COMPONENTS



×5

PUSH BUTTONS

What it does: Complete a circuit while pressed.

How it works: Pressing the cap joins two metal contacts inside, so electricity can pass.

Handle withcare: Place across themiddle gap of the breadboard. Turn it a quarter turn if it seems always on.



×1

BUZZER

What it does: Turns electricity into sound.
How it works: A tiny disc inside vibrates quickly when electricity flows, and makes sound.

Handle with care: Mark + on topis the positive leg. Do not cover the sound hole.



×1

POTENTIOMETER

What it does: A knob that changes resistance.

How it works: Turning the knob slides a contact along a resistor, so the path gets easier or harder.

Handlewith care: Usethemiddlelegand one outer leg for the dimmer missions.



1 metre

ALUMINIUMTAPE

What it does: A shiny strip that carries electricity.

How it works: Aluminium is a conductor. A strip of tape can act like a flat wire.

Handlewithcare: Peelslowly.Cut withcare. Keep away from the open battery wires.



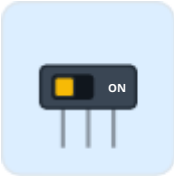
×2

LED RINGS

What it does: Circles of tiny lights.

How it works: Many small LEDs are arranged around one ring and light up together.

Handle withcare: Theyare not programmable. Check the + and – markings before connecting.



×1

SWITCH

What it does: Stays on or off where you leave it.

How it works: Sliding it joins or separates two contacts, like opening and closing a gate.

Handlewith care: Usethe middlepinand one outer pin as the on/off connection.



×6

CROCODILE WIRES

What it does: Wires with clips for things that cannot plug in.

How it works: The spring-loaded jaws bite onto metal, so you can clip onto motor tabs and tape.

Handle withcare: Cliponlyontobaremetal. Do not let two bare clips touch.



needed

9V BATTERY (NOT IN KIT)

What it does: Gives your circuit its push of electricity.

How it works: Chemicals inside push electricity out of the + side and pull it back into the – side.

Handle withcare: Ask agrown-up togetone. Always connect LAST. Disconnect when finished.

ELECTRICITY 101

WHAT IS ELECTRICITY?

Everything ismade oftiny particles. Some are called **electrons**. When electrons move along a wire, we call it **electric current**.

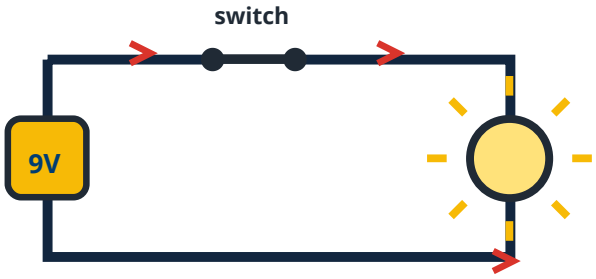
Think of water in a pipe: the battery is the pump, the wire is the pipe and the current is the flow.

WHAT IS A CIRCUIT?

A**circuit** is a loop thatelectricity can travel round. It must start at the battery's + side and finish back at the – side.

If the loop has a break anywhere, the electricity stops.

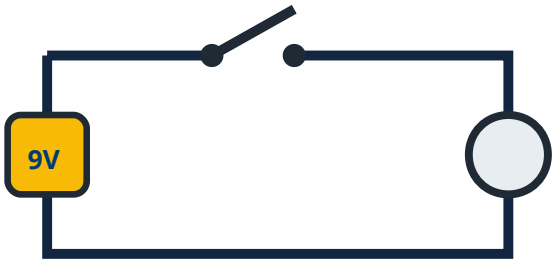
CLOSED CIRCUIT



CLOSED = ELECTRICITY FLOWS

The loop is complete. The lamp glows.

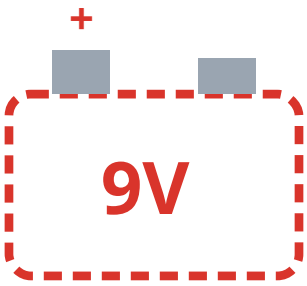
OPEN CIRCUIT



OPEN = NOTHINGFLOWS

There is a gap. The lamp stays dark.

POSITIVE (+) AND NEGATIVE (–)



A battery has two sides. Electricity is pushed out of the **positive (+)** side and returns to the **negative (–)** side.

Some parts care which way round they go: LEDs and buzzers have a + leg and a – leg. Motors just spin the other way.

BATTERY RULE: Never connect the + and – battery wires straight to each other. That is a **short circuit** . It makes wires hot and drains the battery fast.

MORE ELECTRICITY

CONDUCTORS

Let electricity flow easily.



Aluminium tape, the metal ends of wires and the metal legs of parts.

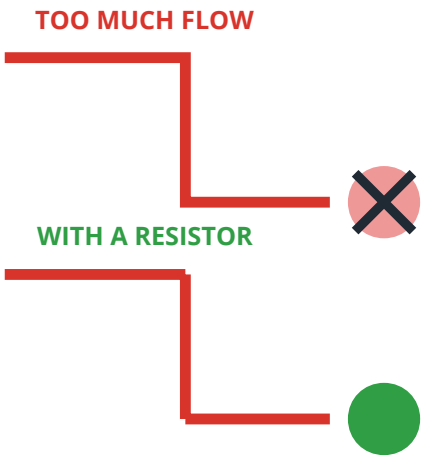
INSULATORS

Stop electricity flowing.



The plastic covering on wires, the breadboard body, the propeller and LED domes.

WHY DO LEDS NEED RESISTORS?



An LED takes as much electricity as it can get, even enough to burn out. A **resistor** acts like a narrow gate that limits the flow.

9V is a strong push, so in this kit each LED uses **two 220Ω resistors in a row**. That keeps the current gentle and safe.

WHAT DO SWITCHES DO?



A switch opens or closes a gap in the circuit. A **push button** is momentary: it works only while pressed. The **slide switch** is maintained: it stays where you put it.

MINI GLOSSARY

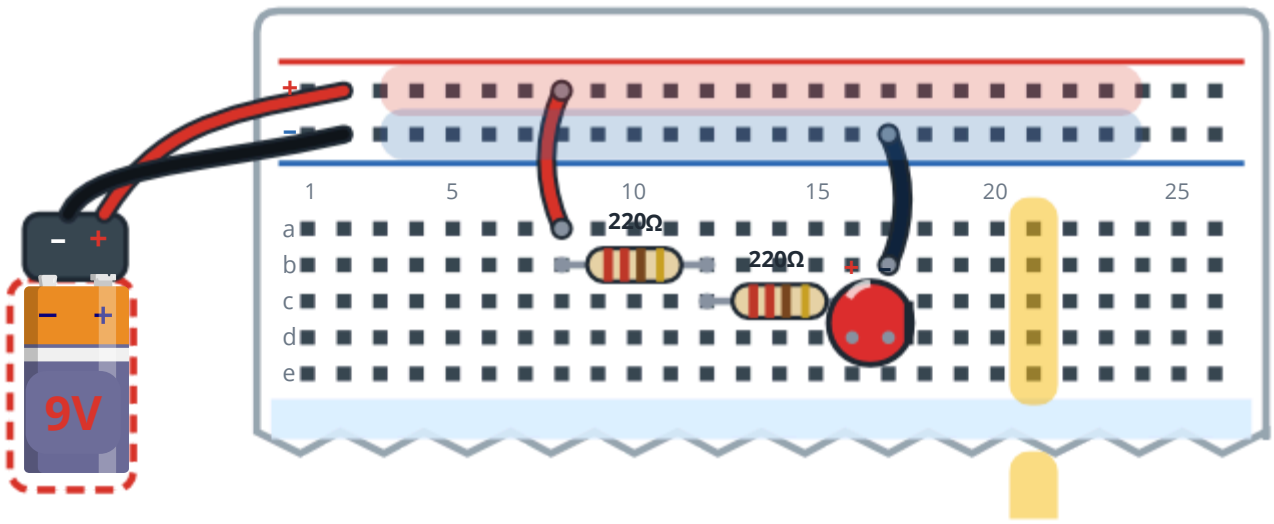
Current: flow of electricity.
Voltage (V): the push that the battery gives.
Resistance (Ω): how hard the flow is made.
Series: parts in a line.
Parallel: parts side by side.

ENGINEER'S RULE

Predict. Build. Test. Fix. Repeat.

Everymistaketeachesyousomething,and that is the engineering process.

YOUR BREADBOARD MAP



Example circuit: red = + supply, dark blue = –, yellow = signal. Yellow glows show holes that are connected.

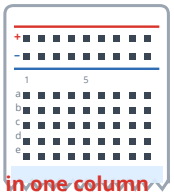
1 · POWER RAILS
The long lines along the edges (red +, blue –) are connected all the way along. Use them for battery power.

2 · COLUMNS
Holesin thesame column are connected in groups of five (yellow). Different columns are NOT connected.

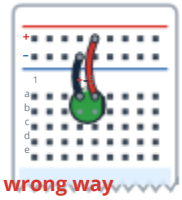
3 · MIDDLE GAP
The two halves are separate. Push buttons straddle the gap so their two sides stay apart until pressed.

4 · ROWS AND NUMBERS
Letters a–jname the rowsand numbers name the columns. "Row b, column 8" is one exact hole.

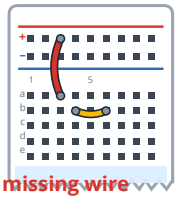
COMMON MISTAKES



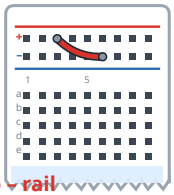
✗ Both legs in one column
The part isshortedout.Spread the legs across different columns.



× LED the wrong way
Long leg (+)mustbeonthe + side. Flip it.

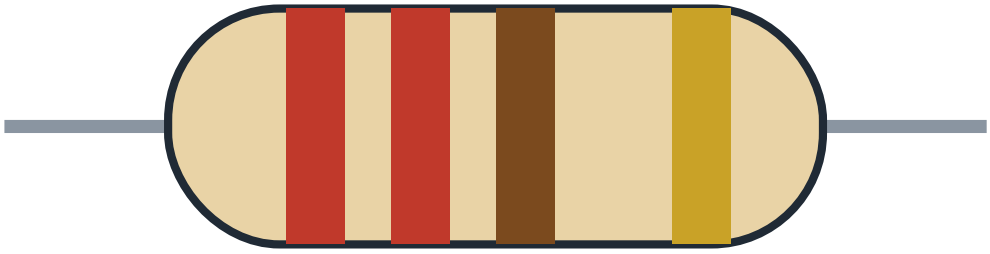


✗ Loose or missing wire
Wire ends mustbepushedin until they stop.



× + wire to – rail
Redwirefrom + to– is a short circuit. Never do this.

RESISTOR COLOUR GUIDE

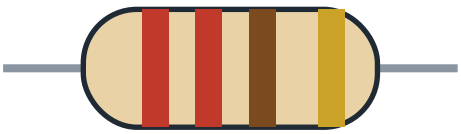


1stdigit 2nddigit multiplier tolerance

COLOUR CODES

Colour	Digit	Multiplier
Black	0	×1
Brown	1	×10
Red	2	×100
Orange	3	×1k
Yellow	4	×10k
Green	5	×100k
Blue	6	×1M
Violet	7	
Grey	8	
White	9	

EXAMPLE: 220Ω



Red (2), red (2), brown (×10) = 22 × 10 = **220 Ω**.

EXAMPLE: 10KΩ



Brown (1), black (0), orange (×1000) = 10 × 1000 = **10,000Ω = 10kΩ**.

DETECTIVE TASK

Pickup allyourresistors. Sort them into two piles by their stripes. How many are in each pile? Write it in your log.

CIRCUIT SYMBOL CHEAT SHEET

Engineers draw circuits with symbols. Every blueprint in this book uses these.



Battery

Wire



Resistor



LED



Push button



Switch



Potentiometer



Buzzer



Motor



LED ring



Tape / conductor



Test gap

WIRE COLOUR CODE USED IN THIS BOOK

positive (+) negative(–) / ground signal (breadboard)

HOW TO READ A BLUEPRINT

1. Find the battery. The long line is +.
2. Follow the red wire out of + through each part.
3. The LED triangle points the way electricity flows.
4. The path must return to the – side, or nothing works.

SOMETHING ISN'T WORKING?

YOUR DEBUGGING SEQUENCE

1. Stop. Unplug the battery.
2. Look at the blueprint and then at your board. Do they match?
3. Check every wire is pushed in all the way.
4. Check LEDs and the buzzer: long leg on the + side.
5. Make sure the circuit is a complete loop from + to –.
6. Change one thing, reconnect the battery and test again.



Mistakes are data. Each one tells you something about how your circuit really works.



LED NOT LIGHTING

Checkthelong leg isonthe + side. Check both resistors are in the circuit.



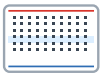
LED INSERTED BACKWARDS

LEDs areone-way.Removethebattery and turn the LED around.



LOOSE JUMPER WIRE

Pull and re-pusheachwireuntil it stops. A wiggly wire is a loose wire.



WRONG BREADBOARD HOLES

Holesin thesame columnarejoined.Different columns are not. Re-check each column number.



SWITCH OR BUTTON NOT WORKING

Turn thebutton a quarterturn. Fortheslideswitch, use the middle pin and one outer pin.



MOTOR NOT SPINNING

Check the crocodileclipstouch bare metal. Check the battery. Let it run only a short time.



BUZZER NOT SOUNDING

Longlegor+marktothe+side.Check wires and the battery.



INCORRECT POLARITY

Redwire to +rail, blackto –rail. Swap if reversed. Motors just spin the other way.



INCOMPLETE CIRCUIT

Trace a fingerfrom+throughevery part to –. Any gap means no flow.



RGB LED STAYS DARK

Checkthe longestleg isinthe common column. Some RGB LEDs may need it connected the other way: ask a grown-up.



BUTTON ALWAYS ON

The buttonmay beturnedthe wrong way. Remove and turn it a quarter turn.



BATTERY WIRES GET WARM

Disconnectimmediately.Lookfor+and – touching, then fix before reconnecting.

SECTION 1

CIRCUIT EXPLORERS

Lights, Electricity & Circuits



Complete these five missions to become a

CIRCUIT EXPLORER

01	**THE FIRST LIGHT**	Circuit Builder
02	**TRAFFIC LIGHT CONTROLLER**	Light Master
03	**RGB COLOUR LABORATORY**	Light Master
04	**THE BRIGHTNESS MASTER**	Light Master
05	**THE BUTTON-CONTROLLED LAMP**	Circuit Builder

01

THE FIRST LIGHT

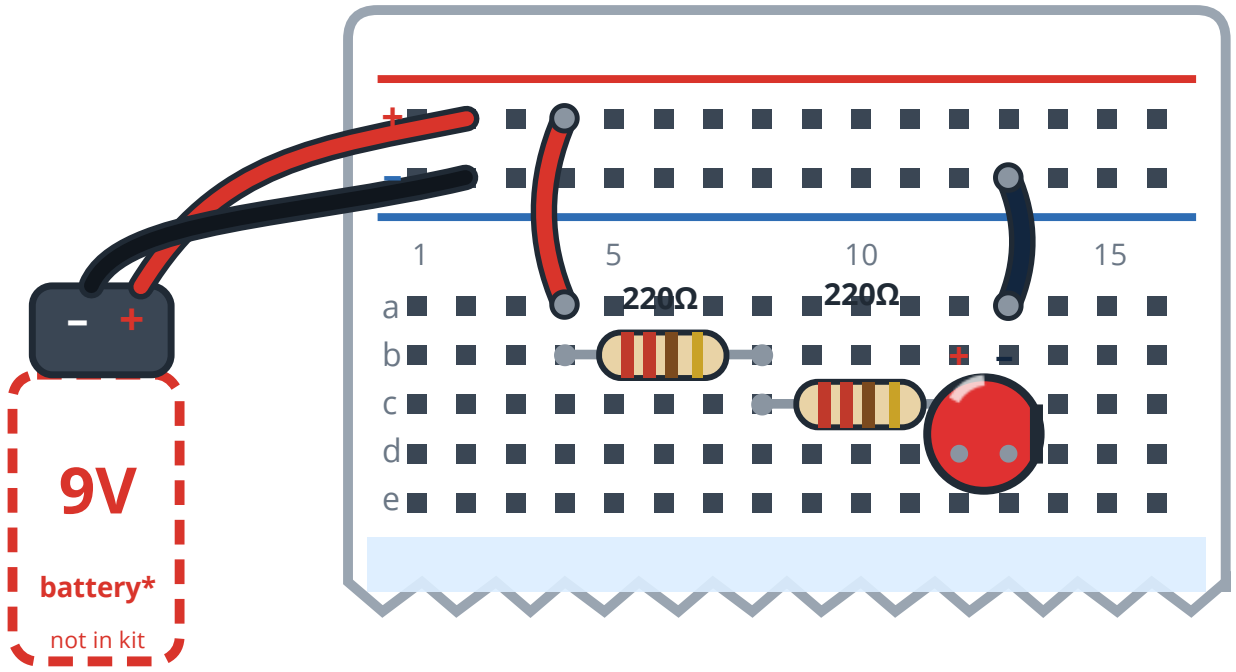
EXPLORER | ENGINEER | INVENTOR

STAMP **CIRCUIT BUILDER** earn it here

MISSION BRIEF

Apower cuthasplunged Mission Control into darkness! You have one LED, two resistors and a battery. Can you bring the first light back on?

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Red LED
- ×2 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How electricity needs a complete loop to flow.
- ★ Why an LED has a + leg (anode) and a – leg (cathode).
- ★ How a resistor protects an LED from too much current.

EXPLORER Follow the instructions and build.

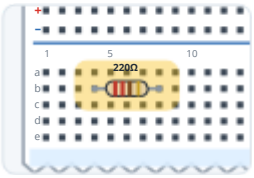
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

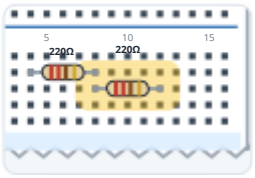
01 THE FIRST LIGHT

BUILD IT

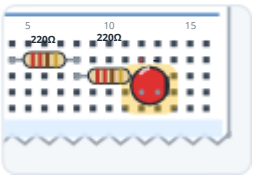
1. Push resistor R1 into row b: one leg in column 4, the other in column 8.



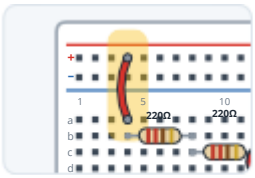
2. Push resistor R2 into row c: one leg in column 8, the other in column 12.



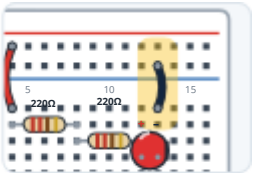
3. Take the red LED. Long leg (+) goes in row d, column 12. Short leg (–) goes in column 13.



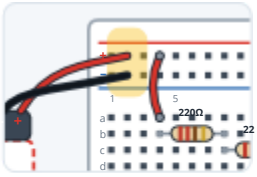
4. Plug a jumper wire from the + rail (red line) to row a, column 4.



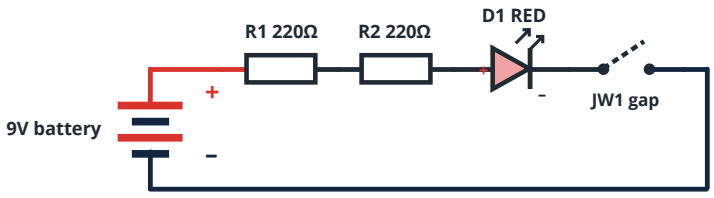
5. Plug a second jumper from row a, column 13 to the – rail (blue line). This is your gap wire.



6. Last: clip the 9V connector onto the battery. Red wire to the + rail, black wire to the – rail. Light!



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Thebattery pusheselectricity outof its + side, through the resistors and the LED, and back into its – side. That loop is a circuit.

When the loop is complete, the LED glows. Pull out the gap wire and the loop is broken, so the light goes out.

Circuit: acomplete loopfor electricity.

Resistor: a part that slows the flow so the LED is safe.

EXPERIMENT ZONE

PREDICT: Whatwill happen when you pull out the gap wire? Try it!

Now swap resistor R2 for the 10kΩ resistor. Is the light brighter or dimmer? Why do you think so?

⚙ ENGINEER CHALLENGE
Swap the orderofR1andR2.Does the LED change? Explain what you see.

★ INVENTOR CHALLENGE
Design a way toswitch thelight on and off without touching the battery or the LED. Test your idea.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The LED is a one-way street. I think this is because...

02

TRAFFIC LIGHT CONTROLLER

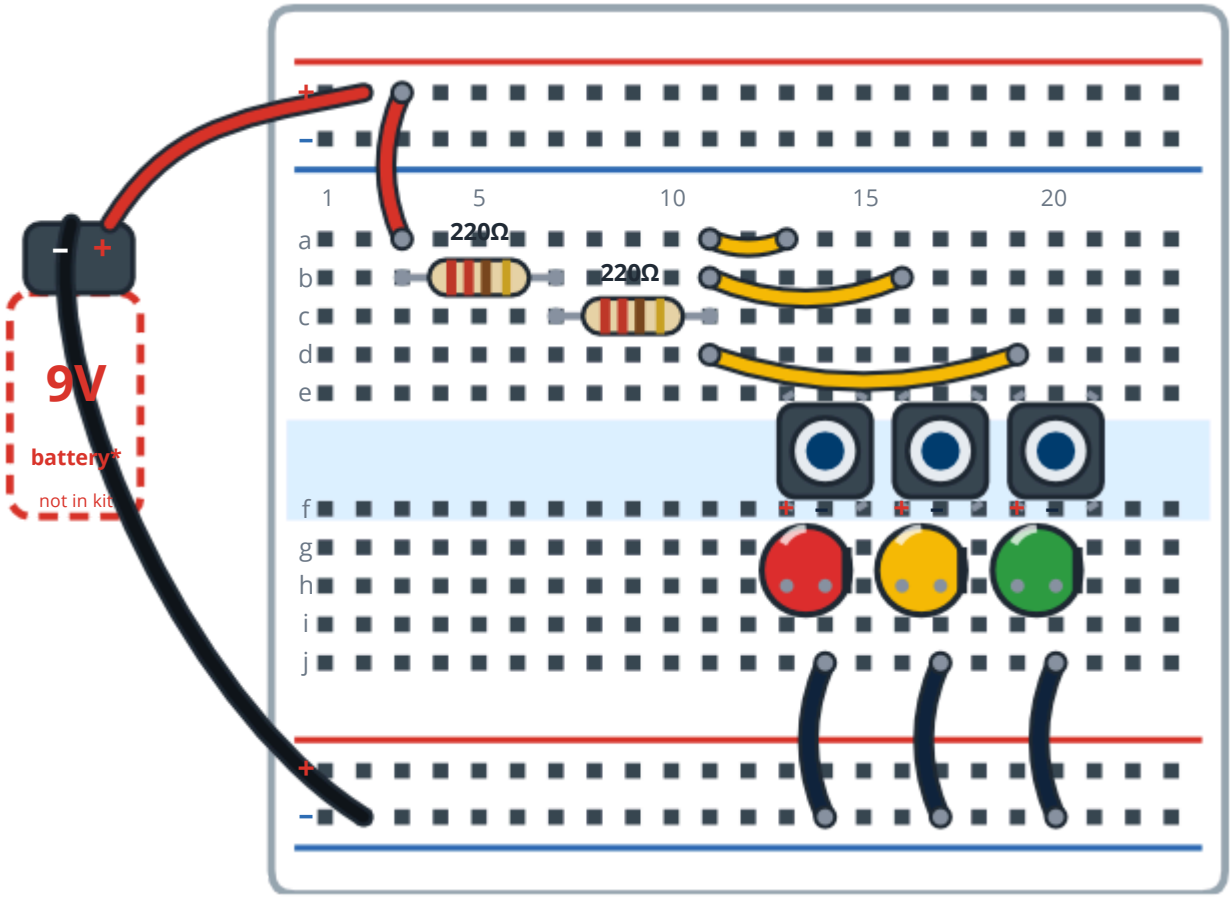
EXPLORER | ENGINEER | INVENTOR

STAMP
LIGHT MASTER
earn it here

MISSION BRIEF

The city traffic lights have broken down and cars are stuck! You are the controller. Build a three-light signal that you operate with your fingertips.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Red LED
- ×1 Yellow LED
- ×1 Green LED
- ×3 Push button
- ×7 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a push button sends electricity down one path.
- ★ Why three lights can share one pair of resistors.
- ★ What a parallel circuit is: paths side by side.

EXPLORER Follow the instructions and build.

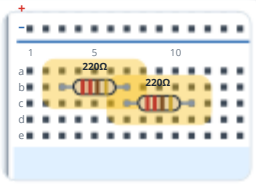
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

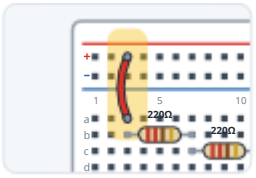
02 TRAFFIC LIGHT CONTROLLER

BUILD IT

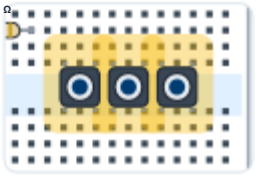
1 Push resistors R1 (row b, columns 3–7) and R2 (row c, columns 7–11).



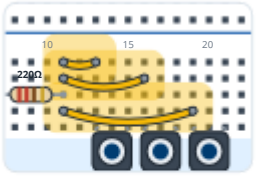
2 Plug a jumper from the + rail to row a, column 3.



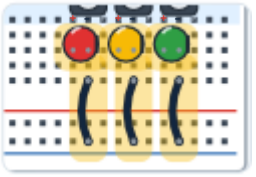
3 Press 3 buttons across the middle gap, at columns 13, 16 and 19.



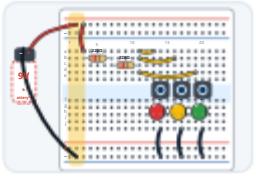
4 Add three jumpers from column 11: to row a column 13, row b column 16 and row d column 19.



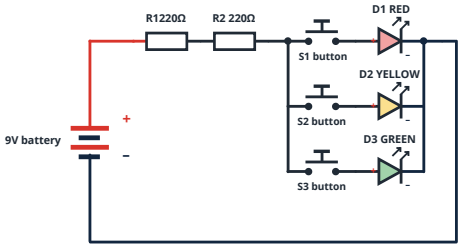
5 Add LEDs in row h: red at 13–14, yellow at 16–17, green at 19–20. Long leg on the left. Jumper each short leg to the bottom – rail.



6 Last: connect the battery. Red to the + rail on top, black to the – rail on the bottom.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Each button is a gate. When you press one, electricity can run down its path and light its LED. The three paths sit side by side, so this is a parallel circuit.

Only one light is usually on, so one pair of resistors protects them all.

Parallel: paths sideby side thatshare thesame battery.

Push button: a switch that works only while you press.

EXPERIMENT ZONE

PREDICT: What happensifyou press two buttons at once?

Try it. Are both lights as bright as before? Write what you notice.

ENGINEER CHALLENGE
Run a real traffic sequence with your buttons: red, then yellow, then green. How long should each light stay on?

★ INVENTOR CHALLENGE
Apedestrianwantsto cross!Design a rule for your buttons so the lights never show a dangerous mix.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

When I pressed two buttons, the lights were...

03

RGB COLOUR LABORATORY

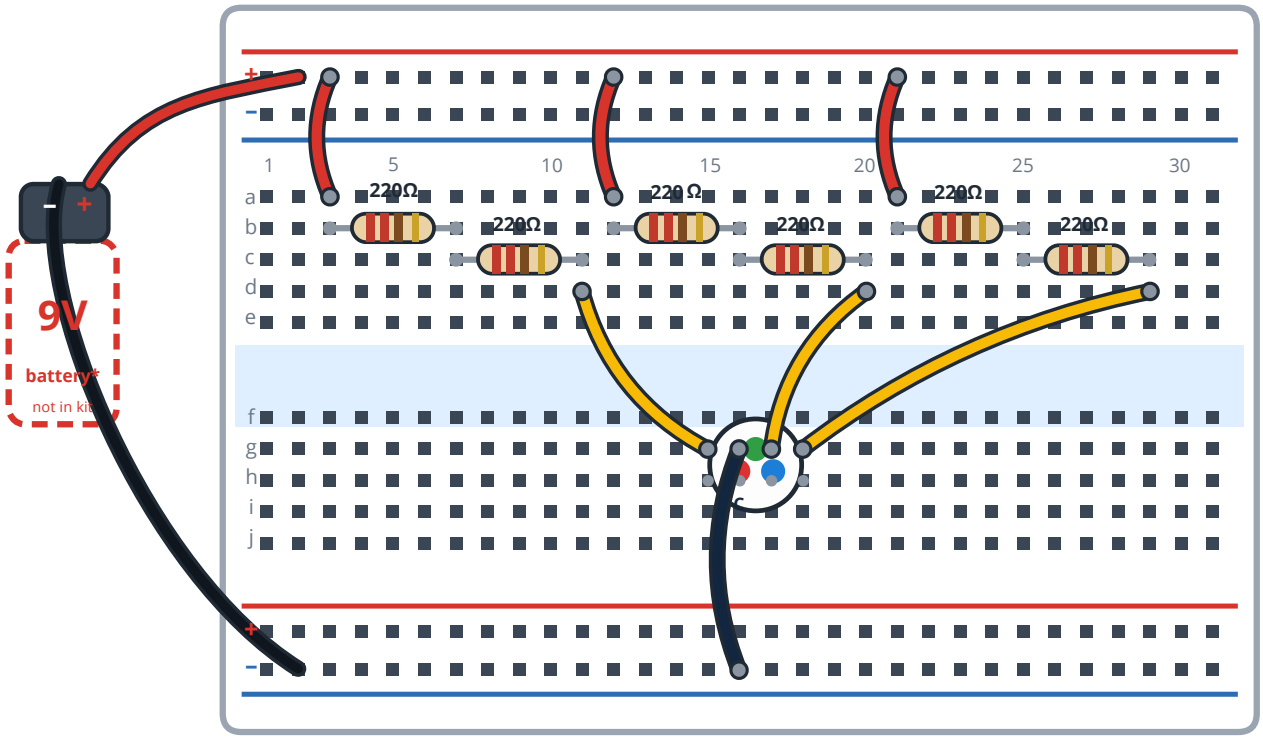
EXPLORER | **ENGINEER** | INVENTOR

STAMP
LIGHT MASTER
earn it here

MISSION BRIEF

TheLight Lab needs a lamp that can glow any colour you ask for. All you have is one tiny RGB LED: red, green and blue light in one bulb. Let the experiments begin!

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×6 220Ω resistor
- ×1 RGB LED
- ×7 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ Red, green and blue light can mix to make new colours.
- ★ Why each colour needs its own resistors.
- ★ How to find out what a part does by testing it.

EXPLORER Follow the instructions and build.

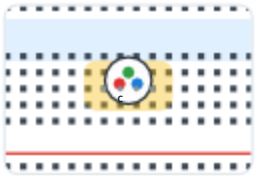
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

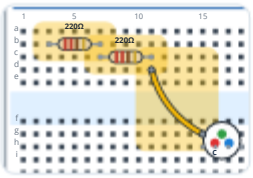
03 RGB COLOUR LABORATORY

BUILD IT

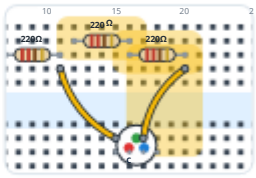
1. Push the RGB LED into row h. The longest leg goes in column 16. The three shorter legs go in columns 15, 17 and 18.



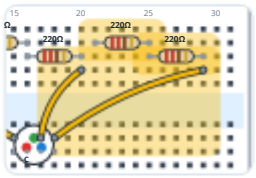
2. Colour line 1: resistors R1 (row b, 3–7) and R2 (row c, 7–11). Jumper from row d, column 11 to row g, column 15.



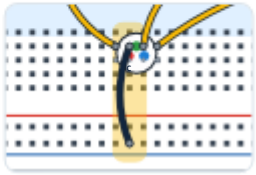
3. Colour line 2: R3 (row b, 12–16) and R4 (row c, 16–20). Jumper from row d, column 20 to row g, column 17.



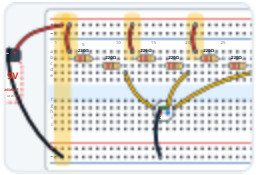
4. Colour line 3: R5 (row b, 21–25) and R6 (row c, 25–29). Jumper from row d, column 29 to row g, column 18.



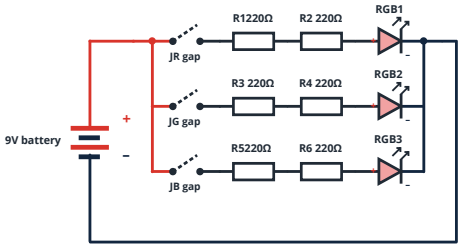
5. Jumper the longest leg (row g, column 16) to the bottom – rail.



6. Connect the battery. Then plug in the three colour wires (+ rail to row a at columns 3, 12 and 21), one at a time.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Inside the RGB LED are three tiny LEDs sharing one leg. Each colour has its own path, so you can switch them on separately or together.

Your eyes mix the lights: red and green look yellow, green and blue look cyan, and all three look nearly white.

RGB: Red, Green, Blue: the three colours of light.

leg: the one leg shared by all three colours.

EXPERIMENT ZONE

DISCOVER: Plug in colour wire 1 only. Which colour is it? Do the same for wires 2 and 3.

Write the colour next to each wire. Now mix two wires, then all three. Record every colour you make.

ENGINEER CHALLENGE
Make a colourchart: every mix you can create with your three colour wires.

★ INVENTOR CHALLENGE
Design a mood lamp that uses colours for feelings. Which colour will be calm? Which will be exciting?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

Wire 1 = ____ Wire 2 = ____ Wire 3 = ____

04

THE BRIGHTNESS MASTER

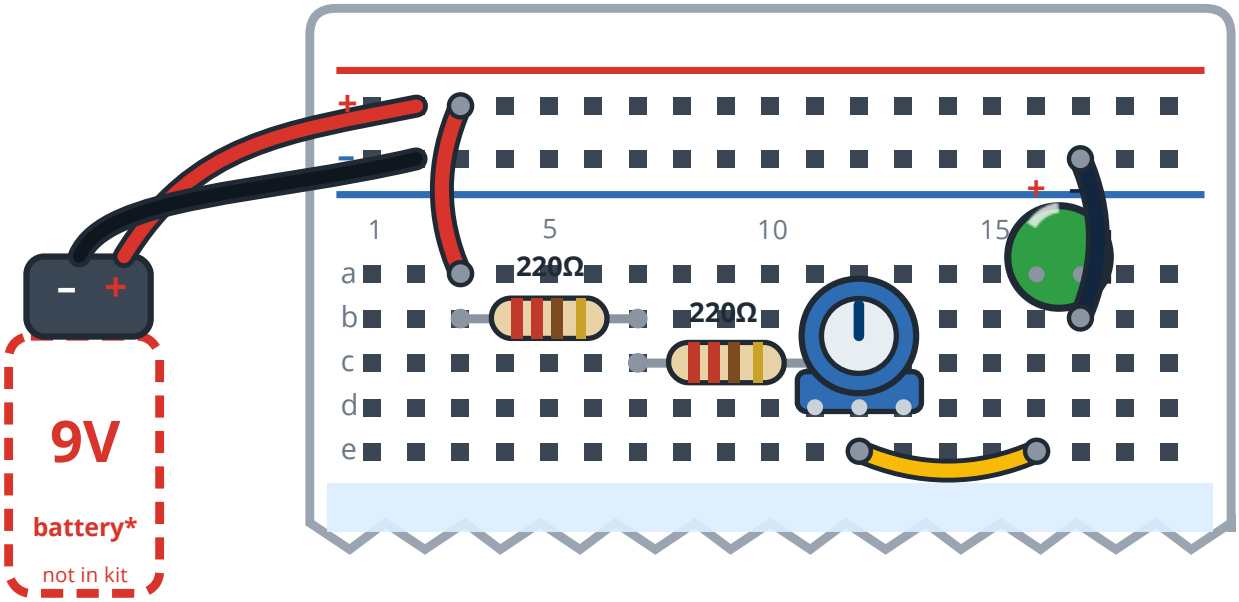
EXPLORER | ENGINEER | INVENTOR

STAMP
LIGHT MASTER
earn it here

MISSION BRIEF

Thelighthouse lampmust shine bright in fog and softly on clear nights. Build a dimmer switch and take control of the light.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Green LED
- ×1 Potentiometer
- ×3 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ What the potentiometer (a knob you can turn) does.
- ★ How more resistance means less current and a dimmer LED.
- ★ How to test a circuit by changing one thing at a time.

EXPLORER Follow the instructions and build.

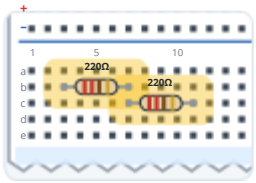
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

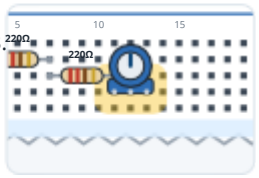
04 THE BRIGHTNESS MASTER

BUILD IT

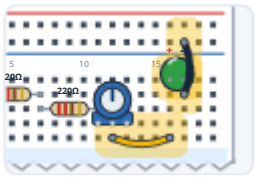
1. Push resistors R1 (row b, 3–7) and R2 (row c, 7–11).



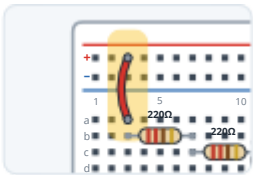
2. Push the potentiometer legs into row d, columns 11, 12 and 13.



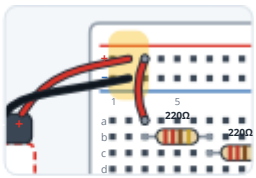
3. Jumper from row e, column 12 to row e, column 16. Add the green LED: long leg row a, column 16, short leg column 17.



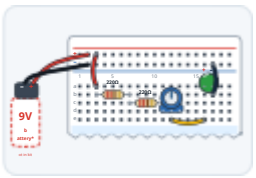
4. Jumper from row b, column 17 to the – rail. Then jumper from the + rail to row a, column 3.



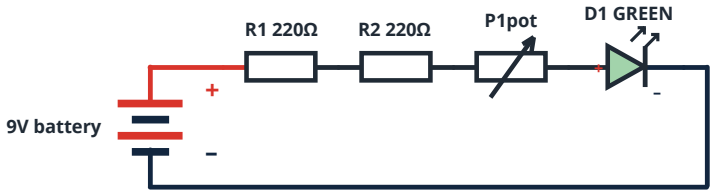
5. Connect the battery: red to the + rail, black to the – rail.



6. Slowly turn the knob from one end to the other. Watch the LED.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The knob changes how hard it is for electricity to pass. Turn it one way and the path is easy: lots of current, a bright LED. Turn it the other way and the path is hard: little current, a dim LED.

The two resistors keep the LED safe even when the knob is at its easiest.

Potentiometer: a resistorwith aknobthatchangesitsresistance.

Current: howmuch electricity flows.

EXPERIMENT ZONE

PREDICT: Will the LEDever go completely dark as you turn the knob?

Test it. Write down where it is brightest, where it is dimmest and what surprised you.

⚙ ENGINEER CHALLENGE

SwapR2 forthe10k Ω resistor.How does the brightest setting change? Why?

★ INVENTOR CHALLENGE

Designa nightlightyou set once and forget. How would you stop little fingers turning it by accident?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The brightest setting was... The dimmest was...

05

THE BUTTON-CONTROLLED LAMP

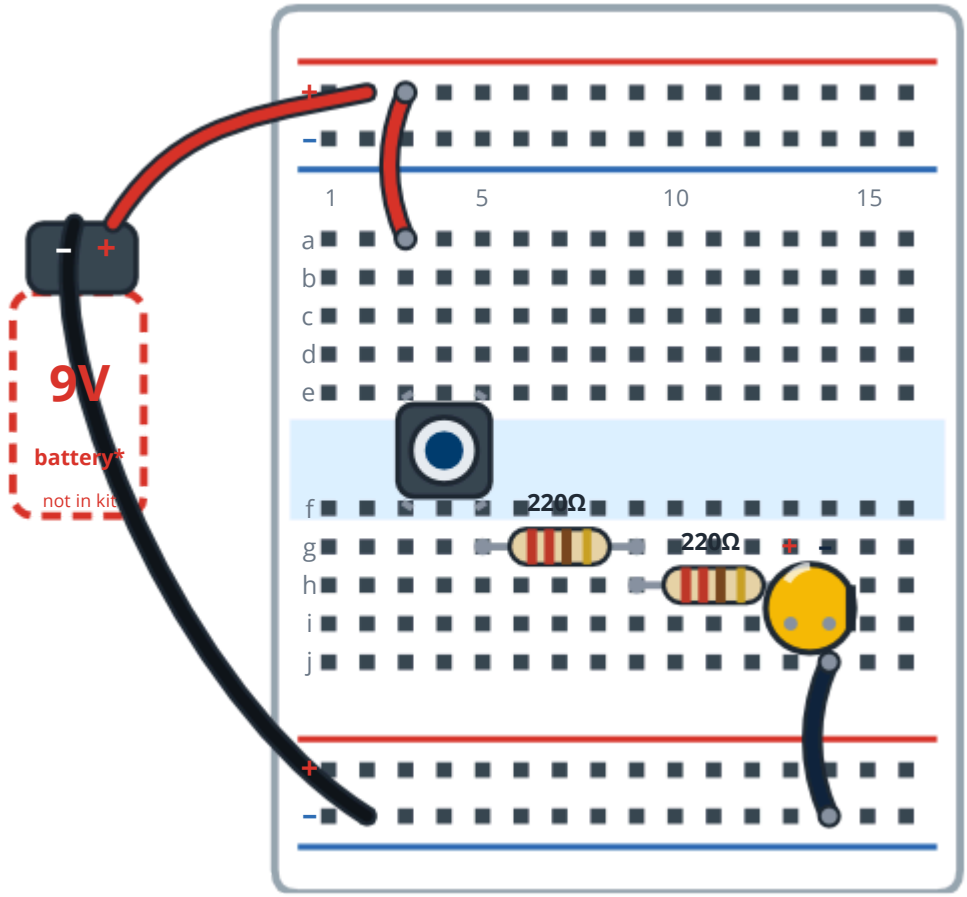
EXPLORER | ENGINEER | INVENTOR

STAMP **CIRCUIT BUILDER** earn it here

MISSION BRIEF

Your robot assistant needs an eye that lights up only when you press its nose. Build a press-to-light lamp and wake the robot!

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Yellow LED
- ×1 Push button
- ×2 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a push button opens and closes a circuit.
- ★ What a momentary switch is.
- ★ How to control a light without touching the wires.

EXPLORER Follow the instructions and build.

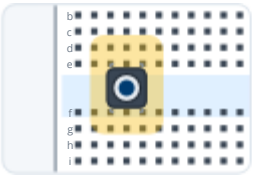
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

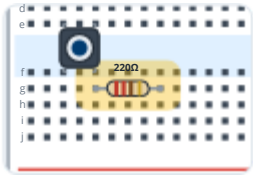
05 THE BUTTON-CONTROLLED LAMP

BUILD IT

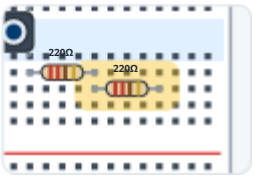
1. Press the button into the breadboard so it sits across the middle gap, at columns 3 to 5.



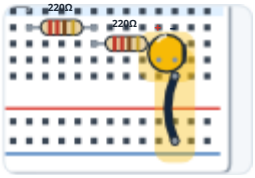
2. Push resistor R1 into row g: columns 5 and 9.



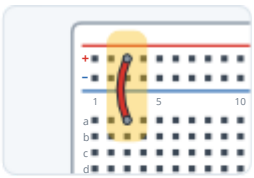
3. Push resistor R2 into row h: columns 9 and 13.



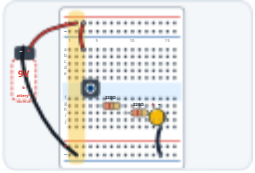
4. Add the yellow LED in row i: long leg column 13, short leg column 14. Jumper from row j, column 14 to the bottom – rail.



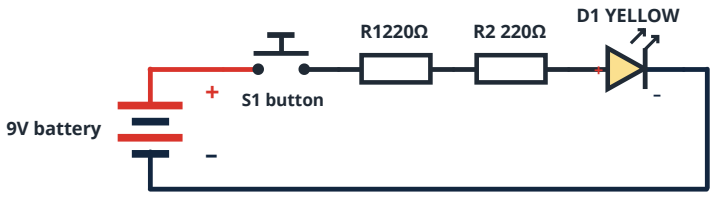
5. Jumper from the + rail on top to row a, column 3.



6. Connect the battery: red to the + rail, black to the – rail. Now press the button!



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Inside the button is atiny gap. When you press, the gap closes and the circuit is complete: the LED lights. Let go and the gap opens again.

A switch that works only while you hold it is called momentary.

Momentary: ononlywhile pressed.

Switch: apartthat opens or closes a circuit.

EXPERIMENT ZONE

PREDICT:Will the LED stay onafter you let go of the button?

Tap it quickly, then hold it for five seconds. What is different?

ENGINEER CHALLENGE
Swap the buttonfortheslideswitch. What is different about how the lamp behaves?

★ INVENTOR CHALLENGE
Invent a doorbelllampforyourbedroom door. Where would the button go?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The button works like a... because...

SECTION 2

INTERACTIVE ENGINEERS

Inputs, Logic & Communication



Complete these five missions to become a

INTERACTIVE ENGINEER

06	**THE TWO-BUTTON SECURITY LOCK**	Logic Detective
07	**THE EITHER-OR SWITCH**	Logic Detective
08	**THE INTRUDER ALARM**	Problem Solver
09	**THE MORSE CODE COMMUNICATOR**	Sound Engineer
10	**THE EMERGENCY BEACON**	Light Master

06

THE TWO-BUTTON SECURITY LOCK

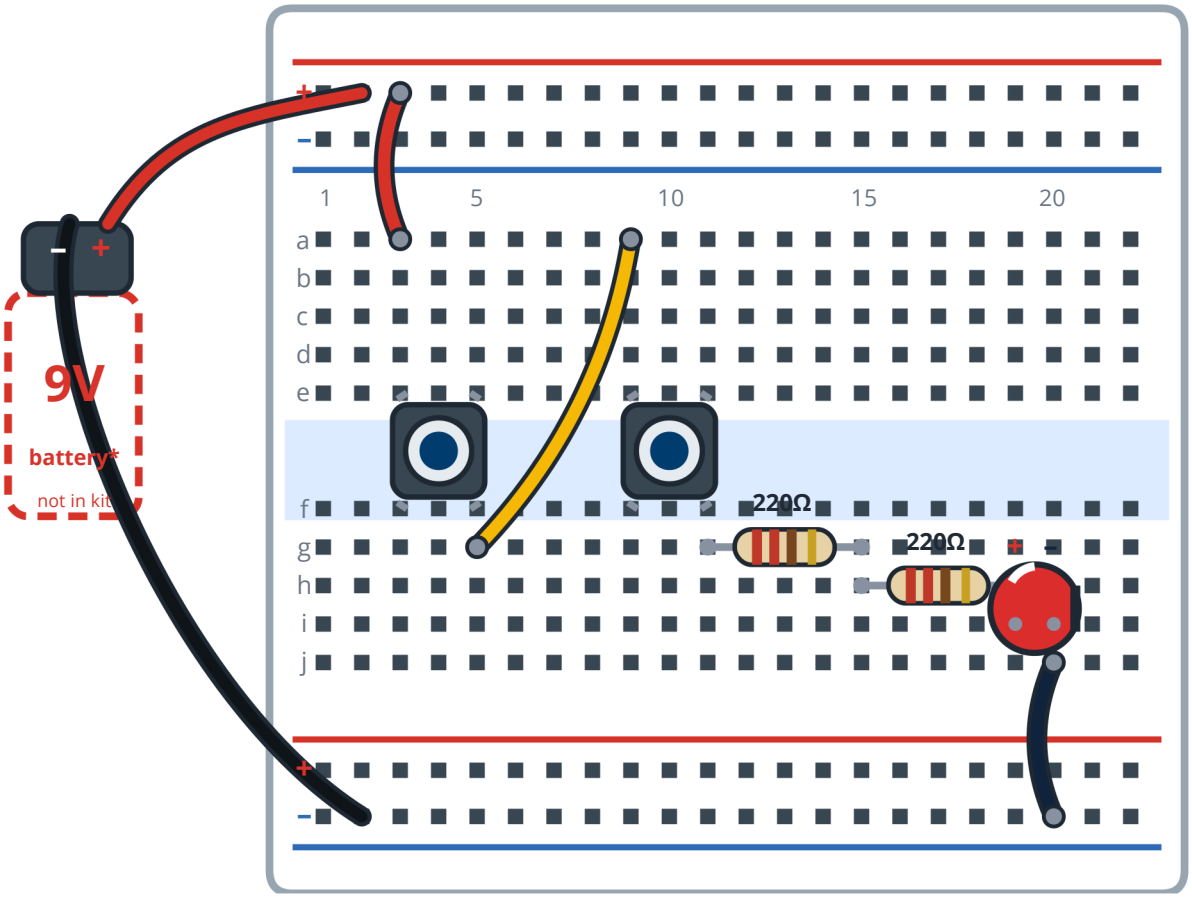
EXPLORER | ENGINEER | INVENTOR

STAMP
LOGIC DETECTIVE
earn it here

MISSION BRIEF

Thevaultat SpaceStation Alpha must only open when two crew members press their buttons at the same time. Build the lock.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Red LED
- ×2 Push button
- ×3 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How two switches in a row make AND logic.
- ★ Why electricity needs every switch closed to flow.
- ★ How to show circuit rules with a truth table.

EXPLORER Follow the instructions and build.

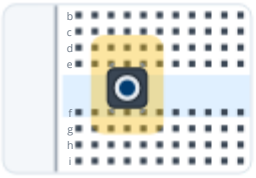
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

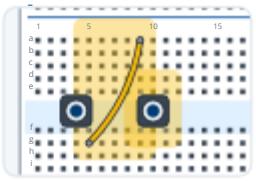
06 THE TWO-BUTTON SECURITY LOCK

BUILD IT

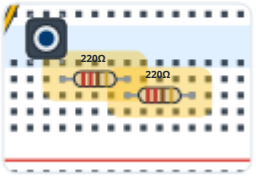
1. Press button S1 across the middle gap at columns 3 to 5.



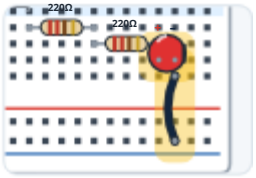
2. Press button S2 across the gap at columns 9 to 11. Jumper from row g, column 5 to row a, column 9.



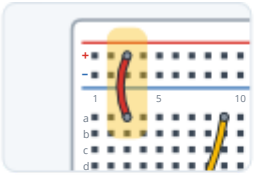
3. Push resistors R1 (row g, 11–15) and R2 (row h, 15–19).



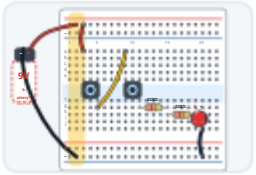
4. Add the red LED in row i: long leg column 19, short leg column 20. Jumper from row j, column 20 to the bottom – rail.



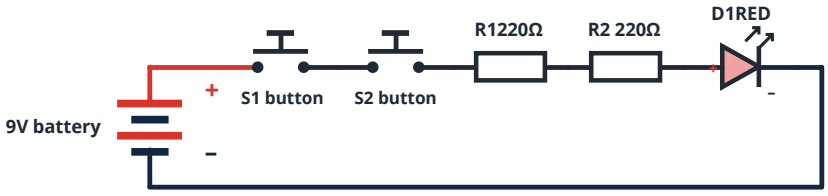
5. Jumper from the + rail on top to row a, column 3.



6. Connect the battery. Test the lock with S1, S2, then both buttons.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The buttons are in series: one after the other on the same path. Electricity must pass through both, so both must be closed. This rule is called AND: the lamp is on only when button 1 AND button 2 are pressed.

Series: partsoneafteranother on onepath.

AND: true only when everything is true.

EXPERIMENT ZONE

PREDICT then TEST. Fill the table with ON or OFF: S1 only, S2 only, both, neither.

Is the light ever on when only one button is pressed? What does this tell you about AND?

S1	S2	LED?
OFF	OFF	
ON	OFF	
OFF	ON	
ON	ON	

⚙ ENGINEER CHALLENGE
Add button S3inthesamerowof the path so three people must press together.

★ INVENTOR CHALLENGE
Design atwo-key systemforatreasure chest where each key holder keeps one button.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The lamp is only on when...

Ω

07

THE EITHER-OR SWITCH

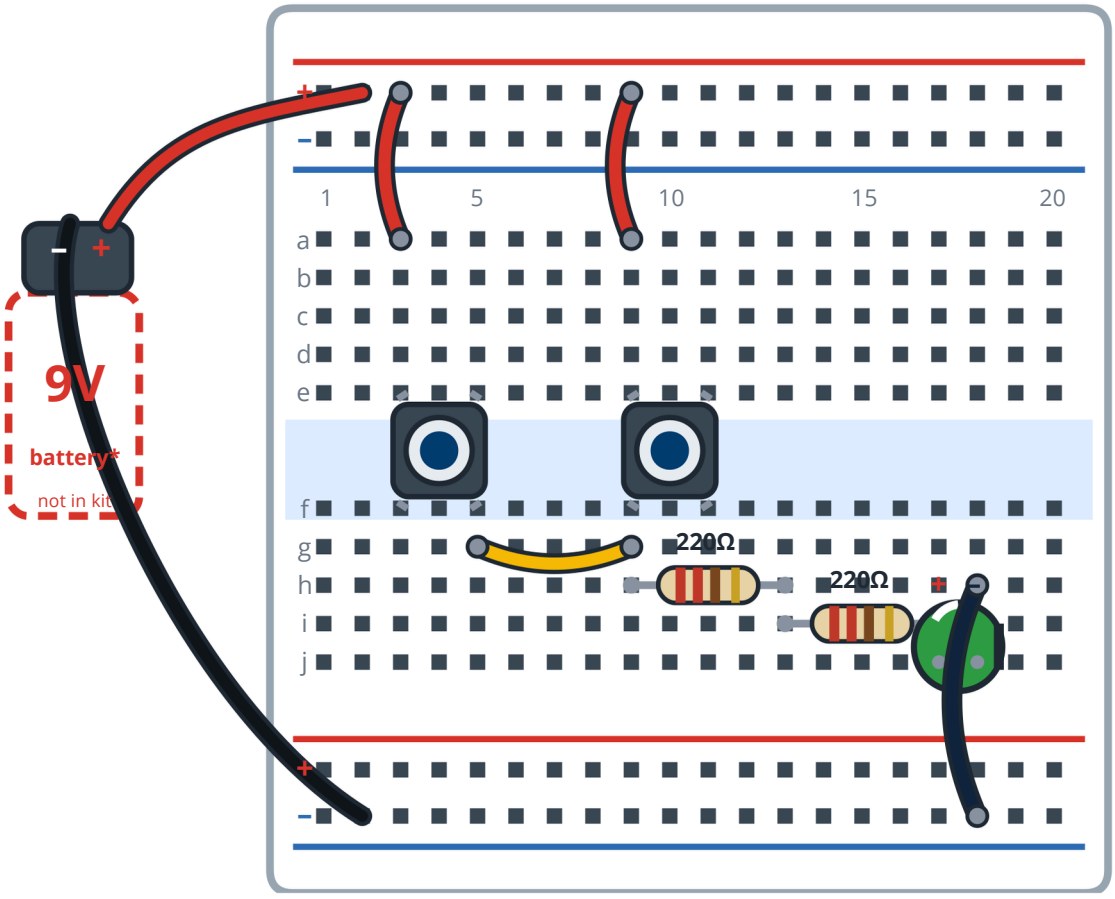
EXPLORER | **ENGINEER** | INVENTOR

STAMP
LOGIC DETECTIVE
earn it here

MISSION BRIEF

Yourspaceshiphas two doors, and a light must come on when either one is opened. Build a circuit that reacts to the front button OR the back button.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Green LED
- ×2 Push button
- ×4 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How two switches side by side make OR logic.
- ★ What a parallel connection does to a circuit.
- ★ How a circuit can have two ways to turn on.

EXPLORER Follow the instructions and build.

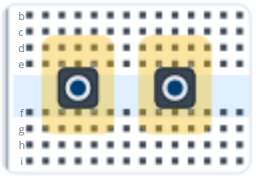
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

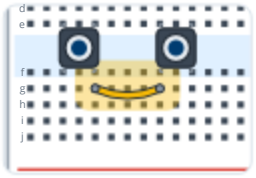
07 THE EITHER-OR SWITCH

BUILD IT

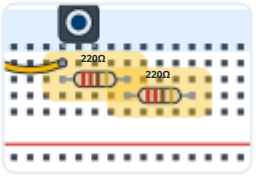
1. Press buttons S1 (columns 3–5) and S2 (columns 9–11) across the middle gap.



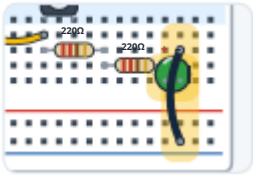
2. Jumper from row g, column 5 to row g, column 9. This joins the two buttons.



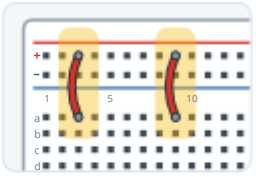
3. Push resistors R1 (row h, 9–13) and R2 (row i, 13–17).



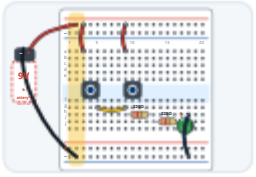
4. Add the green LED in row j: long leg column 17, short leg column 18. Jumper from row h, column 18 to the bottom – rail.



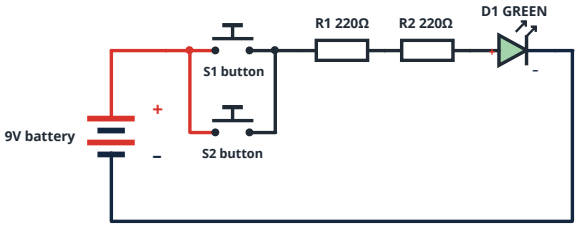
5. Two jumpers from the + rail: to row a column 3 and to row a column 9.



6. Connect the battery. Press S1, then S2, then both.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The buttons sitside by side,so electricity has two possible paths. Either path being closed completes the circuit.

This rule is called OR: the lamp is on when button 1 OR button 2 OR both are pressed.

OR: truewhen atleastone thingistrue.

Parallel: two paths side by side.

EXPERIMENT ZONE

PREDICT: What is differentabout the table for OR compared to AND?

Fill in the table and then compare it with your Mission 6 table.

S1	S2	LED?
OFF	OFF	
ON	OFF	
OFF	ON	
ON	ON	

⚙ ENGINEER CHALLENGE
Add athirdbuttonS3inparallel so the light has three ways to turn on.

★ INVENTOR CHALLENGE
Designahallwaylightwithabutton at each end of the corridor.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

AND is like... OR is like...

08

THE INTRUDER ALARM

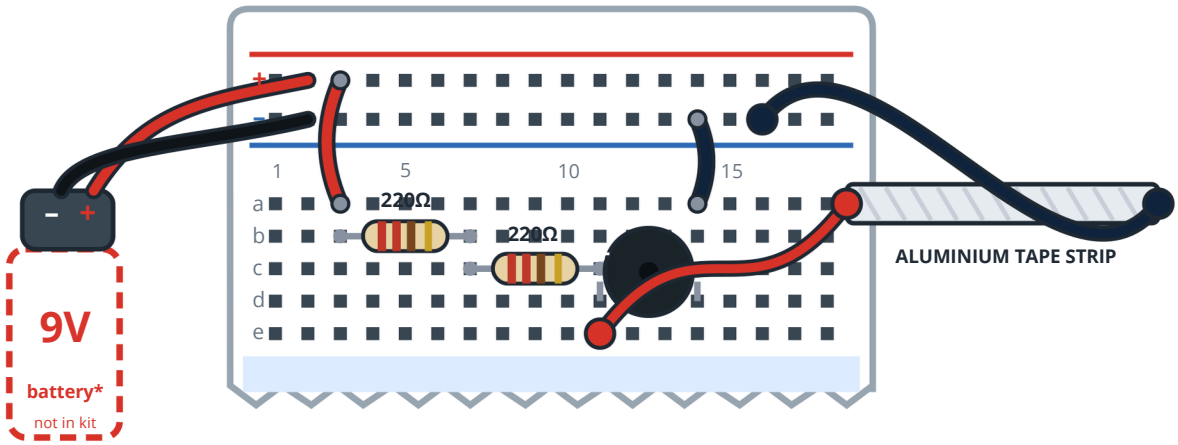
EXPLORER | ENGINEER | **INVENTOR**

STAMP **PROBLEM SOLVER** earn it here

MISSION BRIEF

Athiefissneaking toward the museum! A strip of aluminium tape guards the door. If it breaks, an alarm must sound. Build the tripwire.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

×2 220Ω resistor	×1 Buzzer	×1 Aluminium tape	×4 Jumper wires
×2 Crocodile wires	×1 Breadboard	×1 9V connector	×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How electricity takes the easiest path.
- ★ How a tape strip can short out a buzzer and keep it quiet.
- ★ How breaking a connection can trigger an alarm.

EXPLORER Follow the instructions and build.

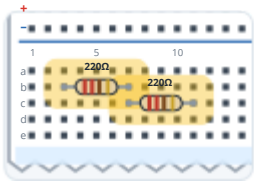
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

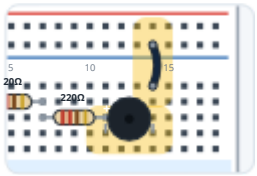
08 THE INTRUDER ALARM

BUILD IT

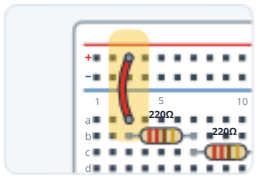
1 Push resistors R1 (row b, 3–7) and R2 (row c, 7–11).



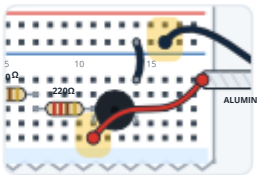
2 Push the buzzer legs into row d: the leg marked + in column 11, the other in column 14. Jumper from row a, column 14 to the – rail.



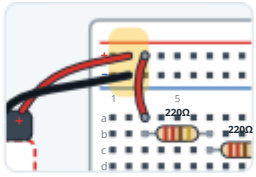
3 Jumper from the + rail to row a, column 3.



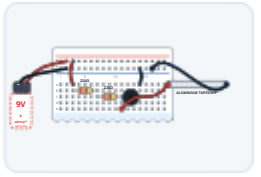
4 Stick a 15cm strip of aluminium tape on the table. Clip crocodile wire 1 to one end and to a jumper in row e, column 11. Clip wire 2 to the other end and to a jumper in the – rail, column 16.



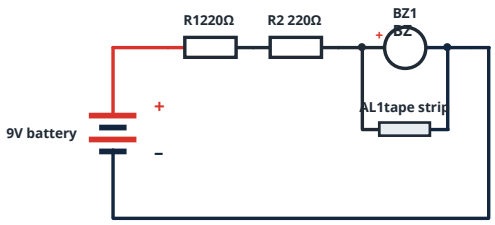
5 Connect the battery. The buzzer stays silent: the tape is carrying the electricity.



6 Be the thief! Lift one clip off the tape. The alarm sounds.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Electricity chooses the easy path. While the tape is joined up, it flows through the tape and skips the buzzer, so the buzzer stays silent.

Break the tape and the electricity must go through the buzzer instead. The alarm sounds.

Conductor: a materialthatletselectricityflow.

Short path: an easy path that skips a part.

EXPERIMENT ZONE

PREDICT: What if you pressthe tape strip back together after the alarm sounds?

Try it. Then try cutting the tape in the middle instead. What do you notice?

ENGINEER CHALLENGE
Make a longertripwirebyclipping two strips together. Does the alarm still work?

★ INVENTOR CHALLENGE
Designatripwireforyourownbedroom door. How could you hide it?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The tape stops the alarm because...

09 THE MORSE CODE COMMUNICATOR

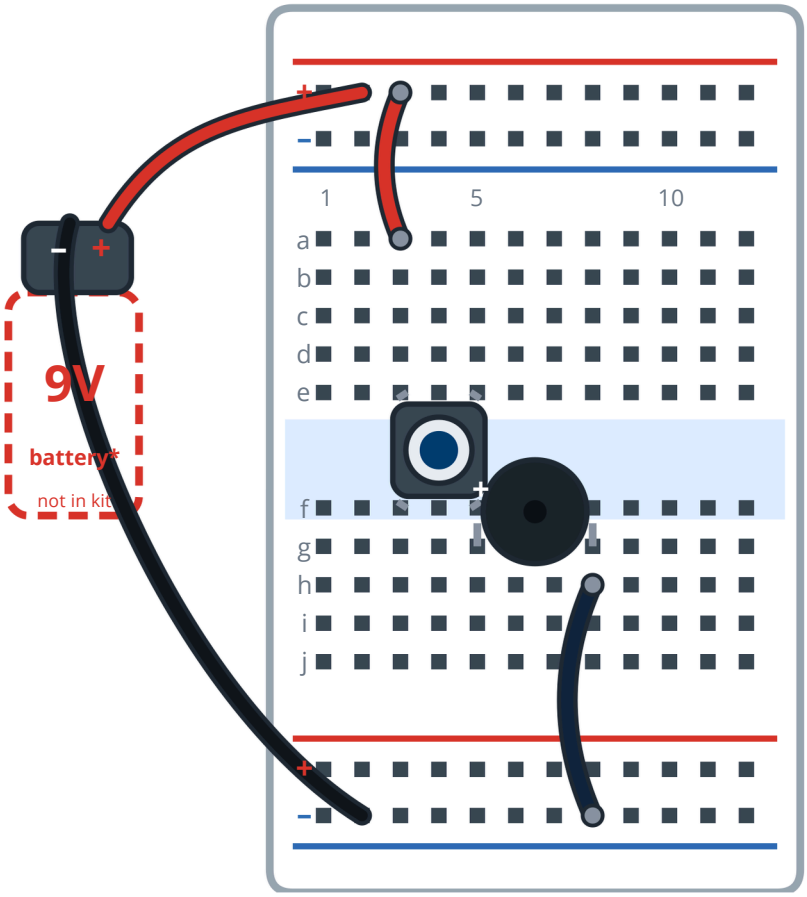
EXPLORER | ENGINEER | INVENTOR

STAMP **SOUND ENGINEER** earn it here

MISSION BRIEF

Yourshipis lostatsea and the radio has failed. The only way to call for help is with short and long beeps. Build a Morse code sender.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×1 Push button
- ×1 Buzzer
- ×2 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a buzzer turns electricity into sound.
- ★ How short and long signals can carry a message.
- ★ How people communicated before telephones.

EXPLORER Follow the instructions and build.

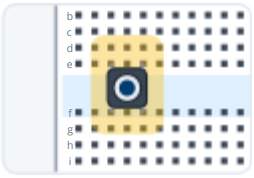
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

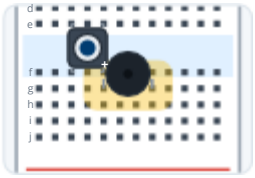
09 THE MORSE CODE COMMUNICATOR

BUILD IT

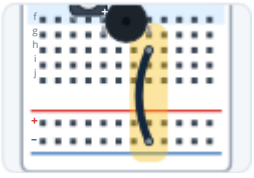
1. Press the button across the middle gap at columns 3 to 5.



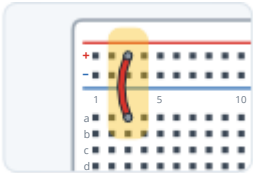
2. Push the buzzer legs into row g: the leg marked + in column 5, the other in column 8.



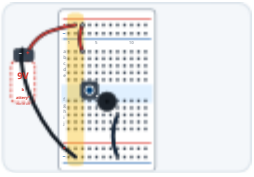
3. Jumper from row h, column 8 to the bottom – rail.



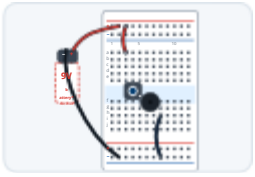
4. Jumper from the + rail on top to row a, column 3.



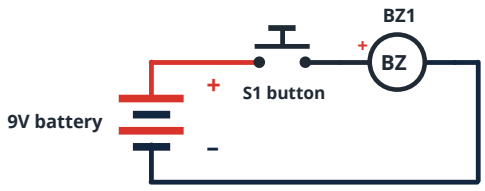
5. Connect the battery: red to the + rail, black to the – rail.



6. Send your first message: tap for a dot (short), hold for a dash (long). Try SOS.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

When electricity flows through the buzzer, a tiny part inside vibrates quickly. The vibrations make sound waves that reach your ears.

The button lets you cut the sound into short dots and long dashes. A dash lasts about three dots.

Buzzer: apart thatturns electricityinto sound.

Morse code: a code of dots and dashes for letters.

A· –	E ·	H· · · ·
I· ·	M– –	N– ·
O– – –	S· · ·	T –

EXPERIMENT ZONE

PREDICT: Cana friendunderstand your message just by listening?

Send your name in Morse code. Ask a friend to write down what they hear.

⚙ ENGINEER CHALLENGE

Send a fullsentence.Timehowlong it takes. How could you speed it up?

★ INVENTOR CHALLENGE

Invent your ownsecretcodewith three different beep lengths. Teach it to a friend.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

My name in Morse code is...

10

THE EMERGENCY BEACON

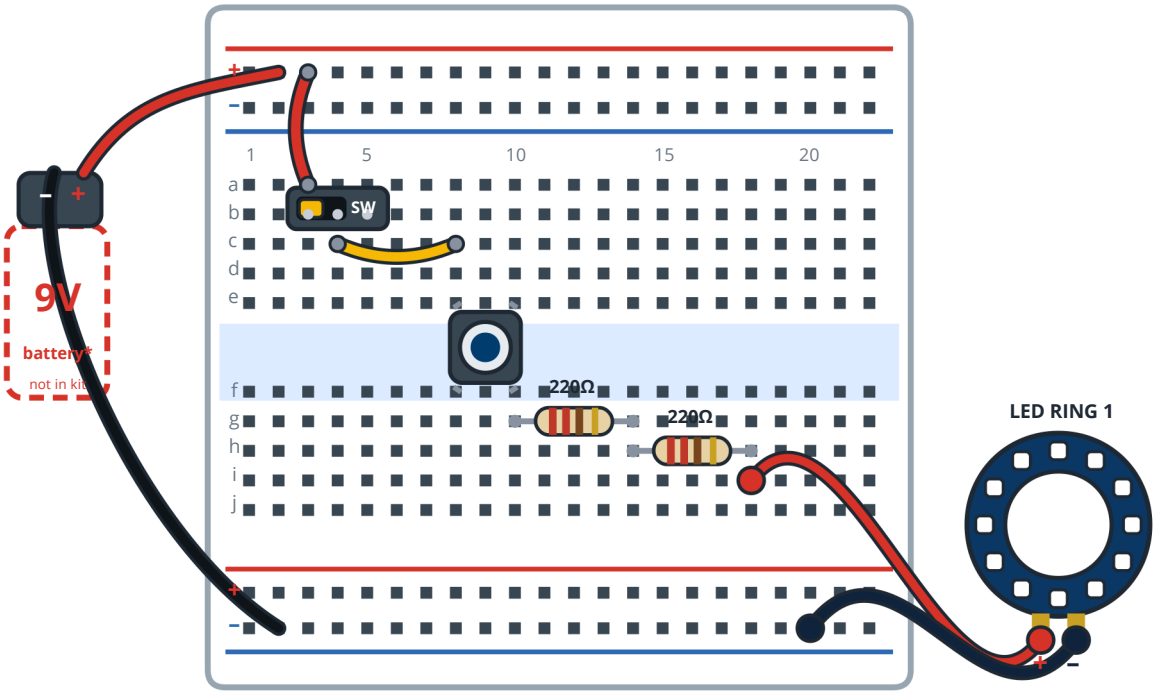
EXPLORER | ENGINEER | INVENTOR

STAMP
LIGHT MASTER
earn it here

MISSION BRIEF

Arescue boatneeds to be seen from far away in the storm. Build a beacon with the LED ring that you arm with a switch and flash with a button.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Push button
- ×1 Slide switch
- ×1 LED ring
- ×4 Jumper wires
- ×2 Crocodile wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a master switch makes a machine safe.
- ★ How flashing patterns carry a message.
- ★ How a switch and a button can work together.

EXPLORER Follow the instructions and build.

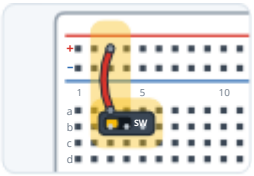
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

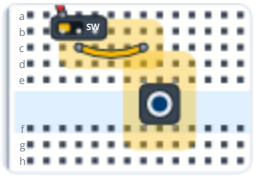
10 THE EMERGENCY BEACON

BUILD IT

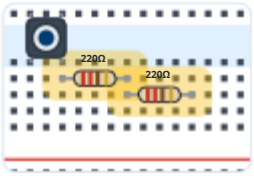
1. Push the slide switch into row b, columns 3 to 5. Jumper from the + rail to row a, column 3.



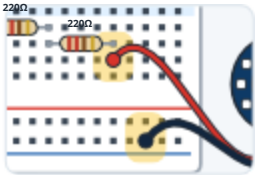
2. Jumper from row c, column 4 to row c, column 8. Press the button across the gap at columns 8 to 10.



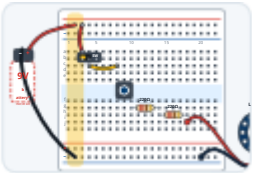
3. Push resistors R1 (row g, 10–14) and R2 (row h, 14–18).



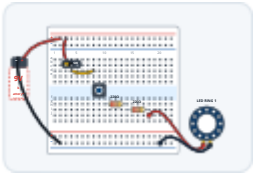
4. Clip crocodile wires to the LED ring: the + one to a jumper in row i, column 18, the – one to a jumper in the bottom – rail, column 20.



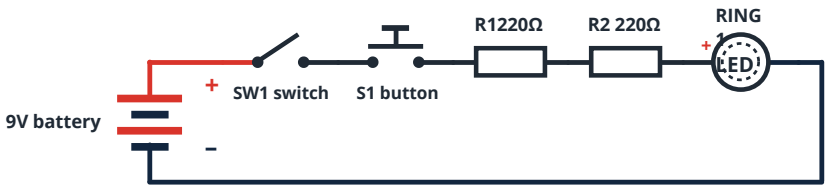
5. Connect the battery: red to the + rail, black to the – rail.



6. Slide the switch to ON to arm the beacon. Tap the button to flash an SOS.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The switch is the master control:nothing works until it is on. The button then decides when the ring flashes.

Flashing grabs attention far better than a steady light, which is why emergency vehicles blink.

Maintained: a switch that stays onor off whereyouleaveit.

Beacon: alight made to be seen from far away.

EXPERIMENT ZONE

PREDICT: What happensif the switch is OFF and you press the button?

Test it. Then arm the switch and flash SOS: three short, three long, three short.

ENGINEER CHALLENGE
Invent a differentflashcodefor a different message, such as HELP or ALL SAFE.

★ INVENTOR CHALLENGE
Design a beacon forabicycle.How would you power it and make it easy to see?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

A steady light means... A flashing light means...

SECTION 3

MOTION MAKERS

Motors, Machines & Energy



Complete these five missions to become a

MOTION MAKER

11	**THE MINI ELECTRIC FAN**	Machine Maker
12	**THE REVERSE ENGINEER**	Machine Maker
13	**THE MACHINE START BUTTON**	Machine Maker
14	**THE WIND POWER LAB**	Problem Solver
15	**THE CONTROL CENTRE**	Machine Maker

11

THE MINI ELECTRIC FAN

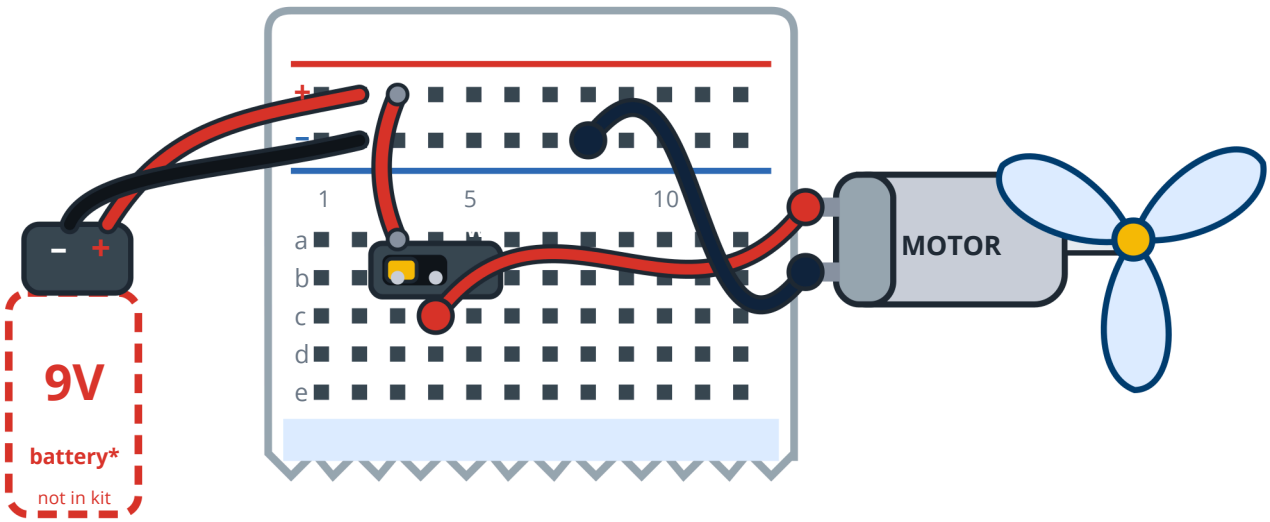
EXPLORER ENGINEER INVENTOR

STAMP
MACHINE MAKER
earn it here

MISSION BRIEF

Itisahotdayintheengine room and the crew is melting. Build a working mini fan to cool everyone down.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×1 Slide switch
- ×1 DC motor
- ×1 Propeller
- ×3 Jumper wires
- ×2 Crocodile wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a motor turns electrical energy into movement.
- ★ How a propeller moves air to make a breeze.
- ★ How a switch controls a machine.

EXPLORER Follow the instructions and build.

ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

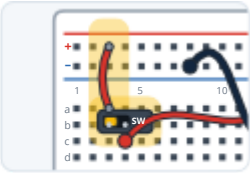
11 THE MINI ELECTRIC FAN

BUILD IT

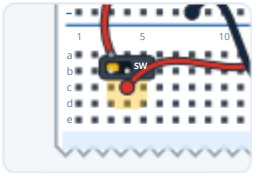
1. Push the propeller onto the motor shaft. Press gently until it sits tight.



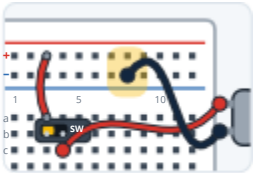
2. Push the slide switch into row b, columns 3 to 5. Jumper from the + rail to row a, column 3.



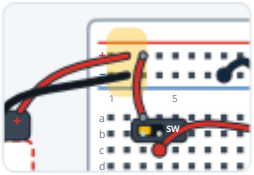
3. Push a jumper into row c, column 4. Clip crocodile wire 1 onto the metal end and the other end onto motor terminal 1.



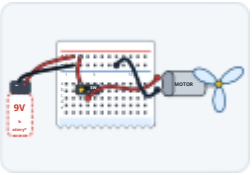
4. Push a jumper into the – rail, column 8. Clip crocodile wire 2 onto it and onto motor terminal 2.



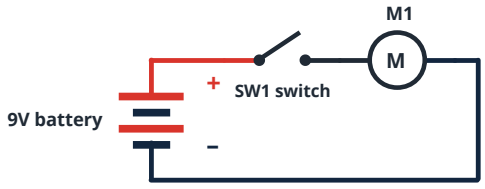
5. Connect the battery: red to the + rail, black to the – rail.



6. Slide the switch to ON. Feel the breeze. Slide it to OFF to stop.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Electricity flowing through the motor makes a magnetic force that spins the shaft. Electrical energy becomes movement.

The propeller pushes air as it spins, so you feel a breeze. That is energy changing form again.

Motor: amachinethatturns electricity intospinning.

Energy: what makes things move, glow or heat up.

EXPERIMENT ZONE

PREDICT: How far away can you feel the breeze?
Hold your hand near the fan, then step back a little at a time. Where does the breeze stop?

⚙ ENGINEER CHALLENGE
Stick a tinyscrapoftapeonone blade. Does it help you see the speed of the propeller?

★ INVENTOR CHALLENGE
Design astand toholdthe fanso it points at your face. What would you use from around your home?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

Energy changes: electricity → ... → ...

12 THE REVERSE ENGINEER

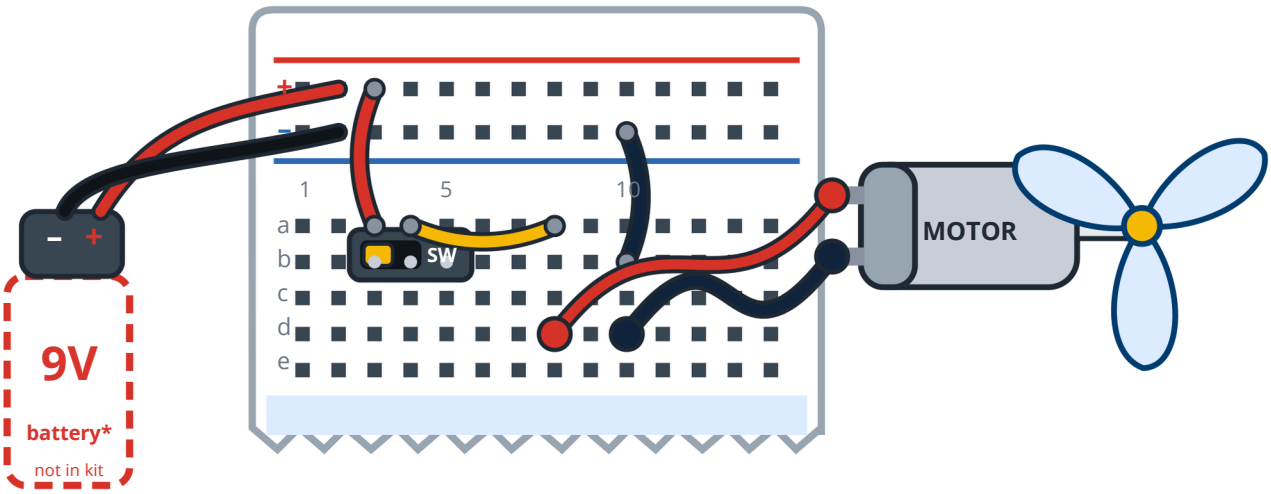
EXPLORER | **ENGINEER** | INVENTOR

STAMP **MACHINE MAKER** earn it here

MISSION BRIEF

The ship's fan blows the wrong way and sucks in hot air! You are the reverse engineer. Find out how to make it spin the other way.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

×1 Slide switch	×1 DC motor	×1 Propeller	×5 Jumper wires
×2 Crocodile wires	×1 Breadboard	×1 9V connector	×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a motor reacts when the electricity flows the other way.
- ★ How to change direction by swapping two connections.
- ★ How engineers test one change at a time.

EXPLORER Follow the instructions and build.

ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

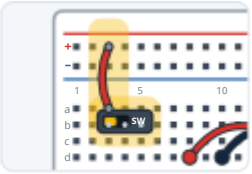
12 THE REVERSE ENGINEER

BUILD IT

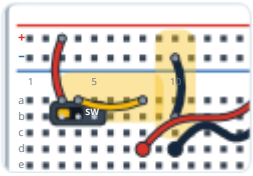
1. Push the propeller onto the motor shaft.



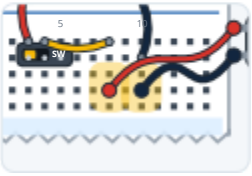
2. Push the slide switch into row b, columns 3 to 5. Jumper from the + rail to row a, column 3.



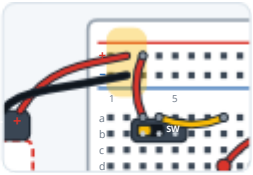
3. Jumper from row a, column 4 to row a, column 8. Jumper from row b, column 10 to the – rail.



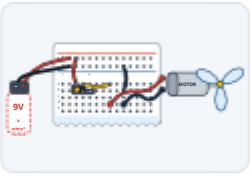
4. Push jumpers into row d, columns 8 and 10. Clip crocodile wire 1 from column 8 to motor terminal 1, and wire 2 from column 10 to terminal 2.



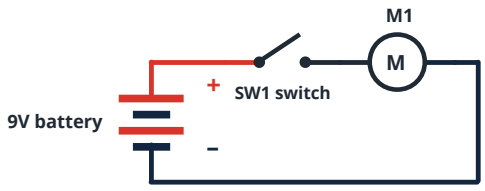
5. Connect the battery. Slide the switch ON. Which way does the fan blow?



6. Switch OFF. Swap the two jumpers: the one in column 8 goes to column 10 and the one in column 10 goes to column 8. Switch ON again.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Amotor spins one way when electricity enters at terminal 1 and the other way when it enters at terminal 2.

You did not change the motor. You only changed which way the electricity flows through it.

Polarity: which side is+and whichsideis–.

Reverse: to go the opposite way.

EXPERIMENT ZONE

PREDICT:Whichway willtheair move after you swap the wires?

Test it. Feel the breeze with your hand. Write down what changed.

ENGINEER CHALLENGE
Find a way tomark the propeller so you can see which way it spins, then record both directions.

★ INVENTOR CHALLENGE
Atoy car mustdriveforwardsand backwards. How could the same trick help?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

Forward airflow was... Reverse airflow was...

13

THE MACHINE START BUTTON

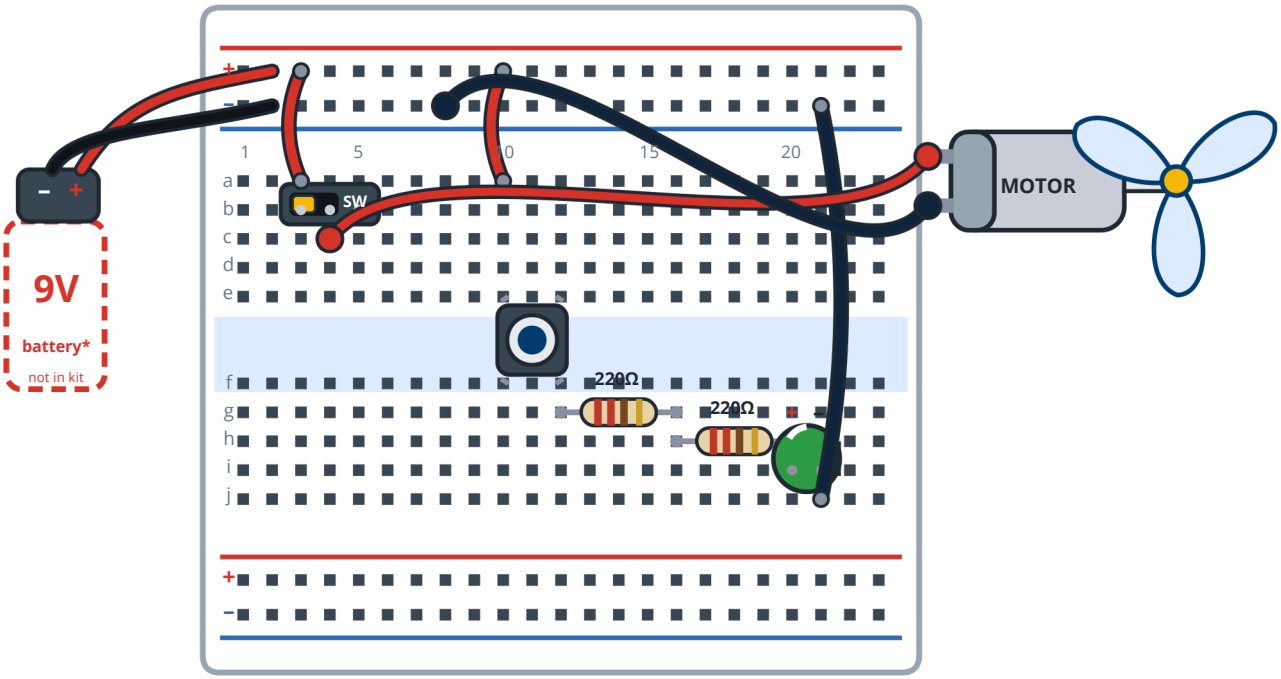
EXPLORER | **ENGINEER** | INVENTOR

STAMP **MACHINE MAKER** earn it here

MISSION BRIEF

Every factory machine has a start control. Build a switch that keeps the fan running and a button that only flashes a start lamp, then compare them.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Green LED
- ×1 Push button
- ×1 Slide switch
- ×1 DC motor
- ×1 Propeller
- ×5 Jumper wires
- ×2 Crocodile wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ The difference between a maintained switch and a momentary button.
- ★ How two separate circuits can share one battery.
- ★ Why engineers choose different controls for different jobs.

EXPLORER Follow the instructions and build.

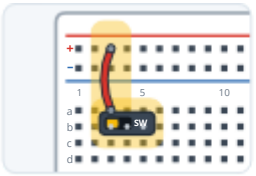
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

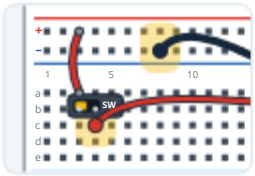
13 THE MACHINE START BUTTON

BUILD IT

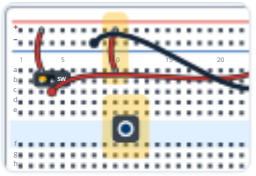
1. Push the slide switch into row b, columns 3 to 5. Jumper from the + rail to row a, column 3.



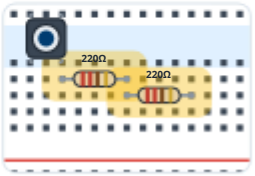
2. Push the propeller onto the motor. Jumper in row c, column 4 and jumper in the – rail, column 8. Clip crocodile wires to the motor terminals.



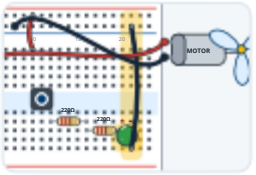
3. Press the button across the gap at columns 10 to 12. Jumper from the + rail to row a, column 10.



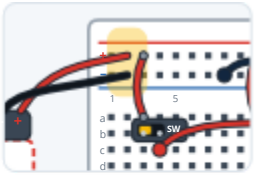
4. Push resistors R1 (row g, 12–16) and R2 (row h, 16–20).



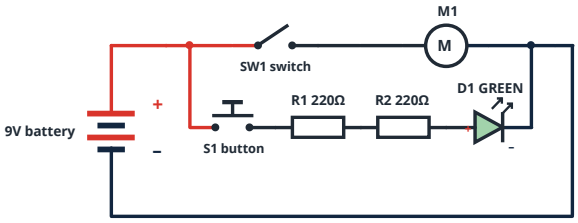
5. Add the green LED in row i: long leg column 20, short leg column 21. Jumper from row j, column 21 to the top – rail.



6. Connect the battery. Slide the switch ON for the fan. Press the button for the lamp.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The slideswitchis maintained: itstays where you leave it. The fan keeps running without your finger.

The button is momentary: the lamp stays on only while you press. Each control suits a different job.

Maintained switch: stays onor offby itself.

Momentary button: works only while pressed.

EXPERIMENT ZONE

PREDICT: Which control would you want on a drill or a power saw, and why?

Run both circuits at once. Which one is easier to stop in an emergency?

⚙ ENGINEER CHALLENGE
Usetheslide switchforthelamp and the button for nothing. What would be different?

★ INVENTOR CHALLENGE
Designacontrol panel foratoyrobot with one momentary and one maintained control. What does each do?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

A switch that stays on is useful for... A button is useful for...

14

THE WIND POWER LAB

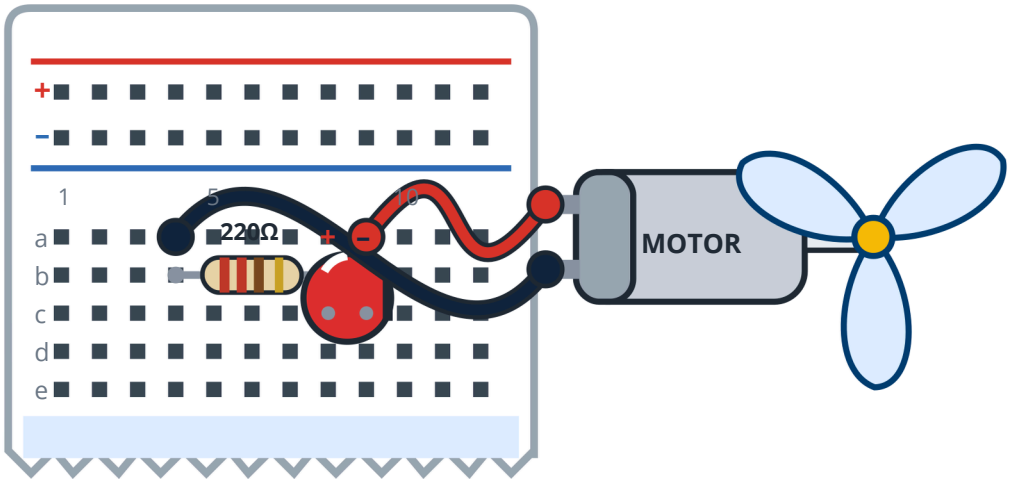
EXPLORER ENGINEER **INVENTOR**

STAMP **PROBLEM SOLVER** earn it here

MISSION BRIEF

A stormis comingand the power is about to fail. Can wind alone make electricity? Turn the motor into a tiny wind generator and light an LED.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×1 220Ω resistor
- ×1 Red LED
- ×1 DC motor
- ×1 Propeller
- ×2 Jumper wires
- ×2 Crocodile wires
- ×1 Breadboard

WHAT WILL YOU DISCOVER?

- ★ How a motor can work backwards as a generator.
- ★ How movement energy turns into electrical energy.
- ★ Why faster spinning makes more electricity.

EXPLORER Follow the instructions and build.

ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

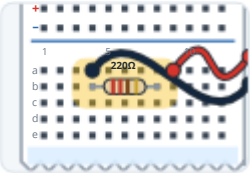
14 THE WIND POWER LAB

BUILD IT

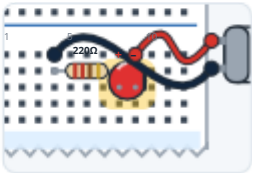
1. Push the propeller onto the motor shaft. No battery this time!



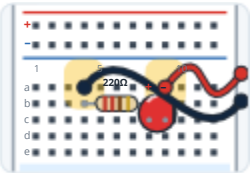
2. Push resistor R1 into row b: columns 4 and 8.



3. Add the red LED: long leg row c, column 8. Short leg column 9.



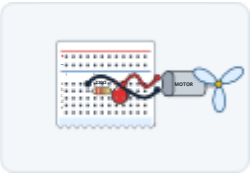
4. Push jumpers into row a, columns 4 and 9. Clip crocodile wire 1 to column 4 and motor terminal 1. Clip wire 2 to column 9 and terminal 2.



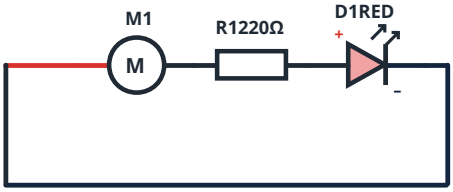
5. Spin the propeller quickly with your finger. Watch the LED.



6. Try blowing on the propeller. Then spin it the other way.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Whenyou spin the motor, wires inside move through magnets and push electricity along. The motor has become a generator. The faster you spin, the more electricity you make. An LED needs the electricity to flow in the right direction, so try spinning both ways.

Generator: amachinethatturnsmovement intoelectricity.

Wind power: electricity made from moving air.

EXPERIMENT ZONE

PREDICT: Will blowinggently or blowing hard make the LED brighter?

Test both. Then run your Mission 11 fan and feel how far its breeze reaches. Write down your findings.

ENGINEER CHALLENGE
Findthespindirectionandspeed that make the LED glow the brightest.

★ INVENTOR CHALLENGE
Avillagehasno electricitybutplenty of wind. Sketch how a wind generator could help.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The LED lit up when... It did not light when...

15

THE CONTROL CENTRE

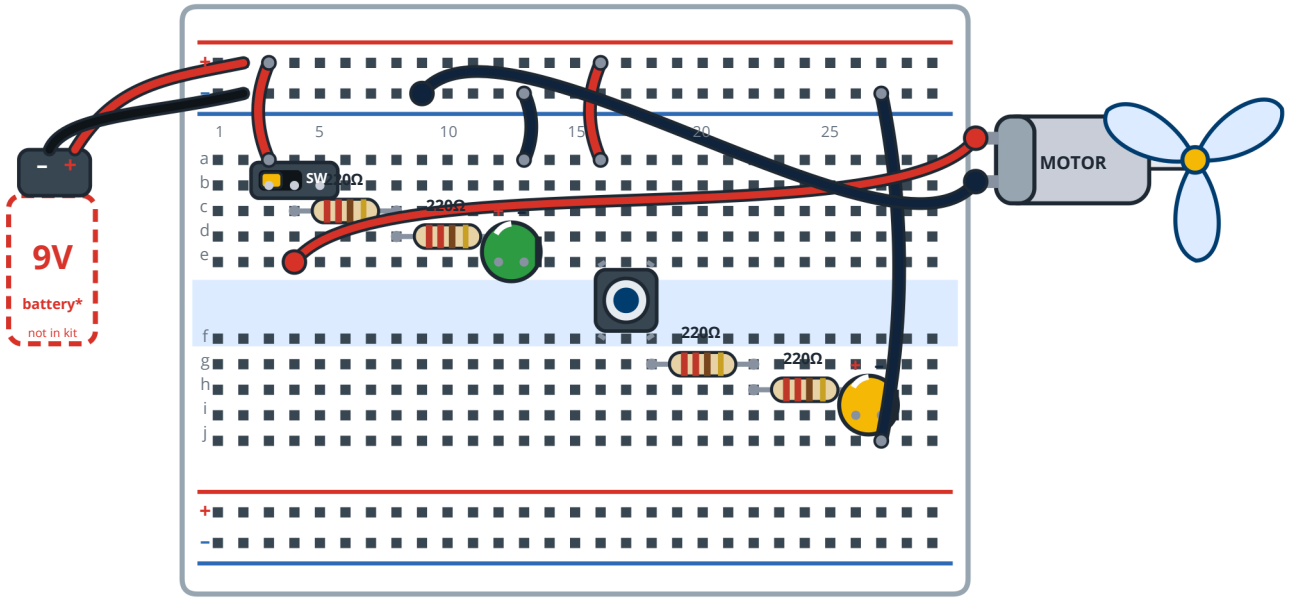
EXPLORER ENGINEER INVENTOR

STAMP
MACHINE MAKER
earn it here

MISSION BRIEF

Youare chiefengineer of a factory machine. Build a control panel with a master switch, a motor, a green POWER ON lamp and a yellow lamp-test button.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

×4 220Ω resistor	×1 Yellow LED	×1 Green LED	×1 Push button
×1 Slide switch	×1 DC motor	×1 Propeller	×6 Jumper wires
×2 Crocodile wires	×1 Breadboard	×1 9V connector	×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How a panel can show the state of a machine.
- ★ How an indicator lamp shares power with a motor.
- ★ Why real machines have a lamp-test button.

EXPLORER Follow the instructions and build.

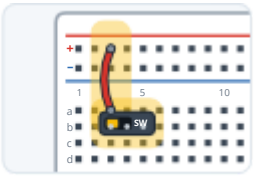
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

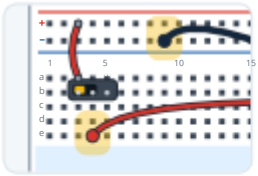
15 THE CONTROL CENTRE

BUILD IT

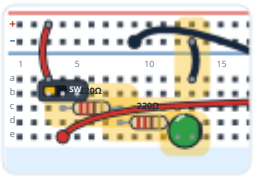
1. Push the slide switch into row b, columns 3 to 5. Jumper from the + rail to row a, column 3.



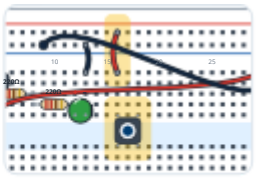
2. Push the propeller onto the motor. Jumper in row e, column 4 and in the – rail, column 9. Clip crocodile wires to the motor terminals.



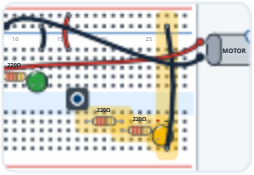
3. POWER ON lamp: resistors R1 (row c, 4–8) and R2 (row d, 8–12), green LED in row e (long leg column 12), jumper from row a, column 13 to the – rail.



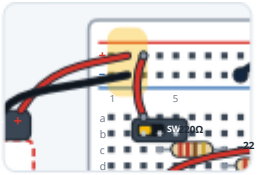
4. Press the lamp-test button across the gap at columns 16 to 18. Jumper from the + rail to row a, column 16.



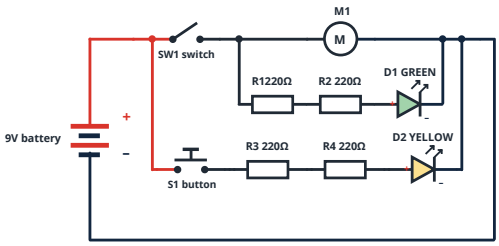
5. Test lamp: R3 (row g, 18–22), R4 (row h, 22–26), yellow LED in row i (long leg column 26), jumper from row j, column 27 to the – rail on top.



6. Connect the battery. Press the test button. Then slide the switch ON.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The master switch sendselectricity to two places at once: the motor and the green lamp. The lamp tells you the machine is on. The test button has its own path, so you can check the yellow lamp works before you start the machine.

Indicator: alight that showswhat amachine is doing.

Control panel: a set of switches and lights for a machine.

EXPERIMENT ZONE

PREDICT: Willthe yellowlampwork when the master switch is OFF?

Test it. Then switch ON and press the button at the same time. What changes?

ENGINEER CHALLENGE
Drawyour controlpanelasalabelled picture. Label every control and light.

★ INVENTOR CHALLENGE
Addarule:themachine must never start unless a safety lamp is on first. How would you wire it?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

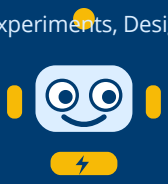
Next time, I would...

The green lamp tells me... The yellow lamp tells me...

SECTION 4

INVENTOR'S LAB

Experiments, Design & Independent Thinking



Complete these five missions to become a

MASTER INVENTOR

16	**THE CONDUCTIVITY DETECTIVE**	Logic Detective
17	**THE LIGHT AND SOUND ALARM**	Sound Engineer
18	**THE DUAL LIGHT DISPLAY**	Light Master
19	**THE SOUND EXPLORER**	Sound Engineer
20	**THE ULTIMATE INVENTOR CHALLENGE**	Creative Inventor

16

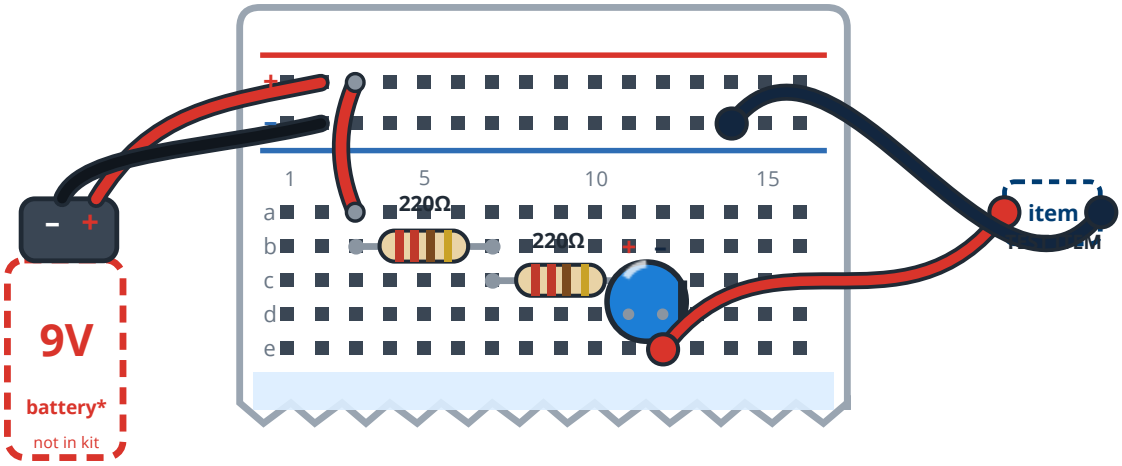
THE CONDUCTIVITY DETECTIVE

EXPLORER | ENGINEER | INVENTOR

MISSION BRIEF

Something in the lab is carrying electricity and you must find out what! Build a detective tester and test every suspect in your kit.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×2 220Ω resistor
- ×1 Blue LED
- ×3 Jumper wires
- ×2 Crocodile wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ The difference between conductors and insulators.
- ★ How a gap in a circuit can be closed by the right material.
- ★ How to run a fair test and record results.

EXPLORER Follow the instructions and build.

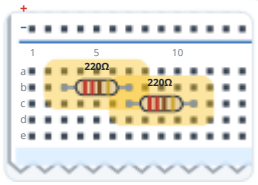
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

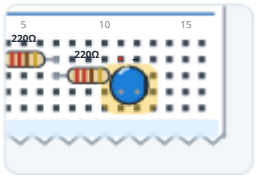
16 THE CONDUCTIVITY DETECTIVE

BUILD IT

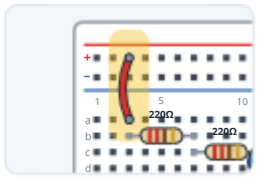
1. Push resistors R1 (row b, 3–7) and R2 (row c, 7–11).



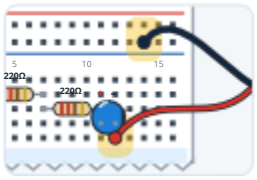
2. Add the blue LED in row d: long leg column 11, short leg column 12.



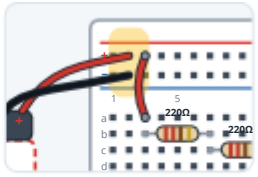
3. Jumper from the + rail to row a, column 3.



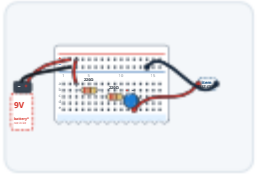
4. Push jumpers into row e, column 12 and into the – rail, column 14. Clip a crocodile wire to each. The free clips are your test probes.



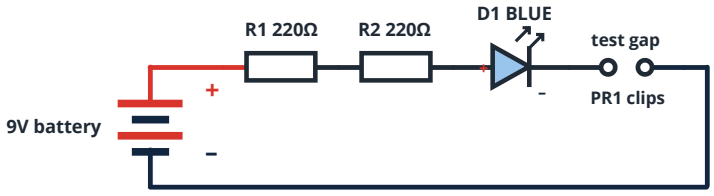
5. Connect the battery: red to the + rail, black to the – rail.



6. Clip the probes across each suspect in the table below. Does the LED light? Write the result.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Thetwoclips leave a gapinthe circuit. If you bridge the gap with a conductor, electricity flows and the LED lights.

If you bridge it with an insulator, nothing flows and the LED stays dark.

Conductor: lets electricity flow.

Insulator: stops electricity flowing.

EXPERIMENT ZONE

PREDICT thenTEST each suspect: tape (shiny side), tape (paper backing), a resistor leg, a jumper wire's plastic, a jumper wire's metal end.

Write YES or NO for each in your log. Which results surprised you?

Suspect	Predict	Result
Tape, shiny side		
Tape, paper backing		
Resistor leg		
Jumper wire plastic		
Jumper wire metal end		

⚙ ENGINEER CHALLENGE
Findoutifthe tape conductswhen it is folded, creased or stuck down. What do you find?

★ INVENTOR CHALLENGE
Design aquizgame whereplayers must find the one conductor among four hidden suspects.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The best conductor was... The best insulator was...

17

THE LIGHT AND SOUND ALARM

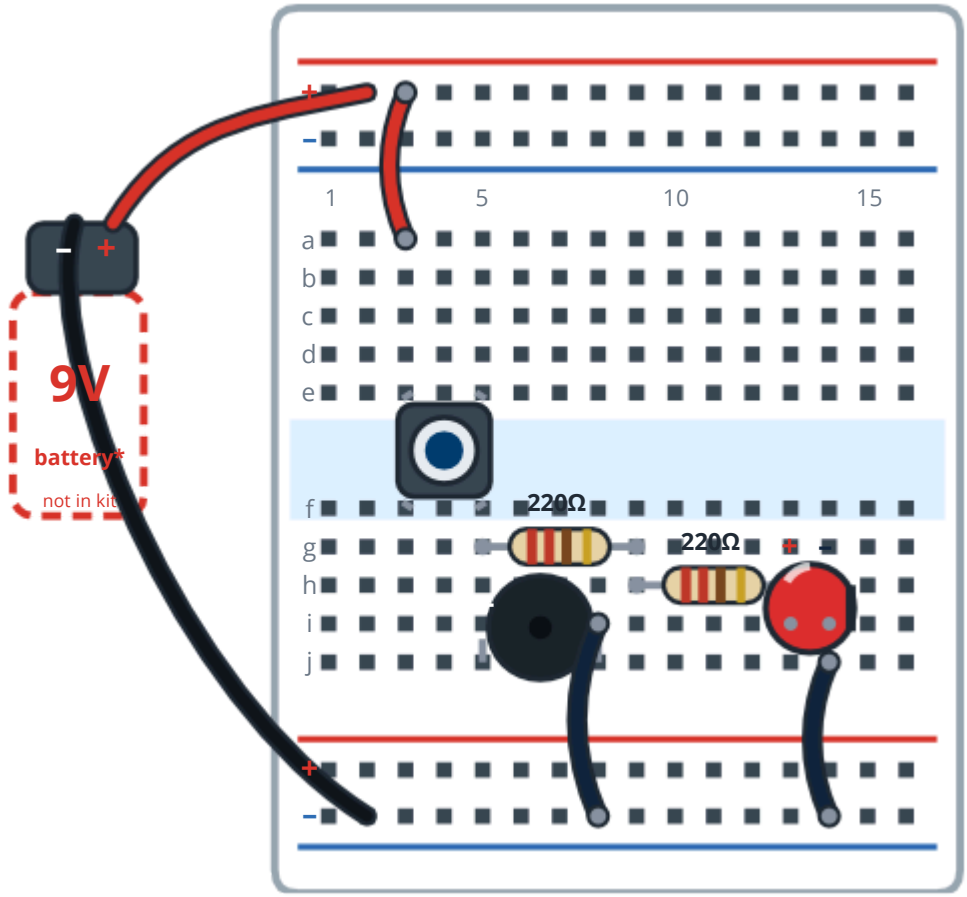
EXPLORER | **ENGINEER** | INVENTOR

STAMP **SOUND ENGINEER** earn it here

MISSION BRIEF

Awarning thatyou only see can be missed, and one that you only hear can be missed too. Build an alarm that flashes AND beeps.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

×2 220Ω resistor	×1 Red LED	×1 Push button	×1 Buzzer
×3 Jumper wires	×1 Breadboard	×1 9V connector	×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How one button can trigger two outputs.
- ★ How light and sound warnings work together.
- ★ How electricity splits between two paths.

EXPLORER Follow the instructions and build.

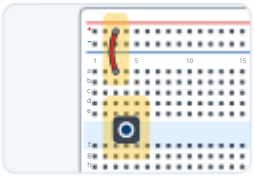
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

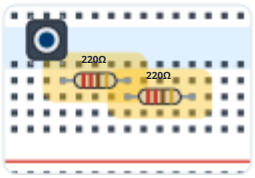
17 THE LIGHT AND SOUND ALARM

BUILD IT

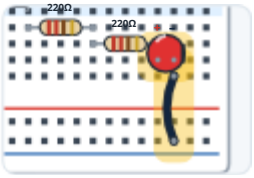
1. Press the button across the gap at columns 3 to 5. Jumper from the + rail to row a, column 3.



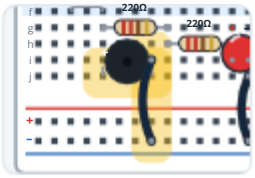
2. Push resistors R1 (row g, 5–9) and R2 (row h, 9–13).



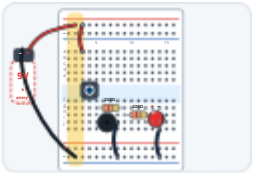
3. Add the red LED in row i: long leg column 13, short leg column 14. Jumper from row j, column 14 to the bottom – rail.



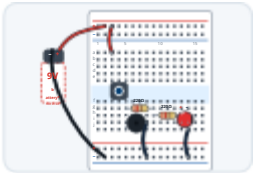
4. Push the buzzer legs into row j: the leg marked + in column 5, the other in column 8. Jumper from row i, column 8 to the bottom – rail.



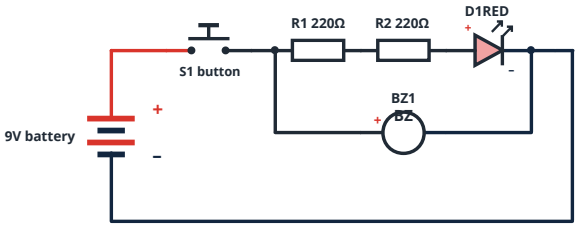
5. Connect the battery: red to the + rail, black to the – rail.



6. Press the button. The light and the sound start together.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

One presssends electricity into two paths at once: one through the resistors and the LED, the other through the buzzer. Warning systems use light and sound together so the message gets through, even if someone is not looking.

Output: apartthat does something,likelightorsound.

Input: apart you use to control a circuit.

EXPERIMENT ZONE

PREDICT: Will the LEDstill workif the buzzer is not connected?

Try pulling out the buzzer's jumper while you press. What happens to the LED?

⚙ ENGINEER CHALLENGE
Swap the redLED fortheblueone. Does the colour change how urgent the warning feels?

★ INVENTOR CHALLENGE
Design awarning systemforaschool fire drill. Which should come first: light or sound?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

The input is... The two outputs are...

18

THE DUAL LIGHT DISPLAY

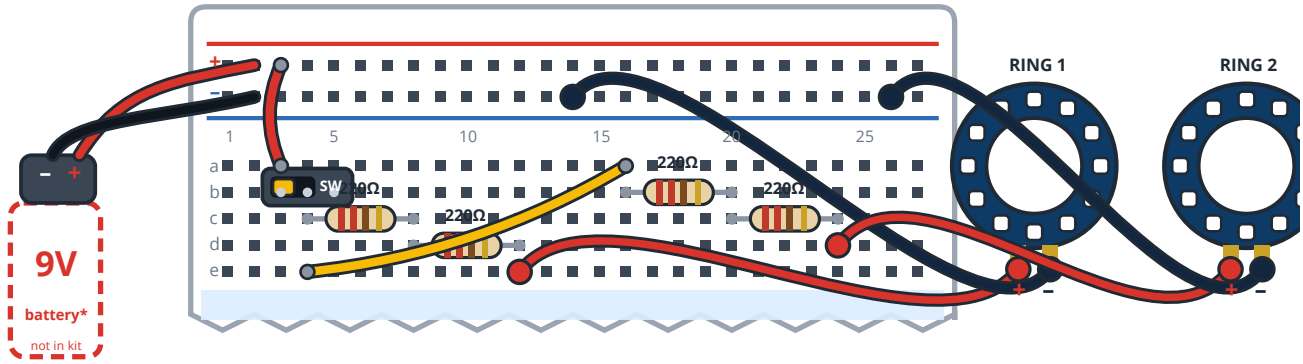
EXPLORER ENGINEER **INVENTOR**

STAMP
LIGHT MASTER
earn it here

MISSION BRIEF

The museum needsaglowing display with two light rings. Build an installation that powers both rings from a single switch.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

×4 220Ω resistor	×1 Slide switch	×2 LED ring	×6 Jumper wires
×4 Crocodile wires	×1 Breadboard	×1 9V connector	×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How one switch can control two separate lights.
- ★ Why each ring has its own pair of resistors.
- ★ How to design a lighting display that looks good.

EXPLORER Follow the instructions and build.

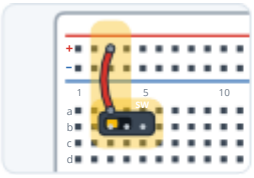
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

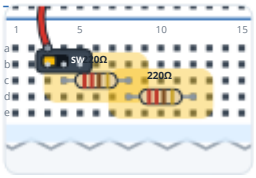
18 THE DUAL LIGHT DISPLAY

BUILD IT

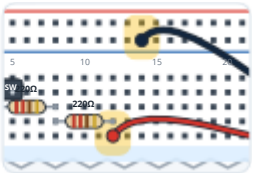
1. Push the slide switch into row b, columns 3 to 5. Jumper from the + rail to row a, column 3.



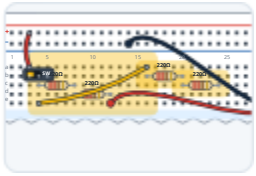
2. Ring 1 resistors: R1 (row c, 4–8) and R2 (row d, 8–12).



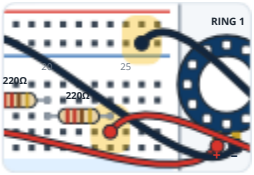
3. Clip crocodile wires to ring 1: the + one to a jumper in row e, column 12, the – one to a jumper in the – rail, column 14.



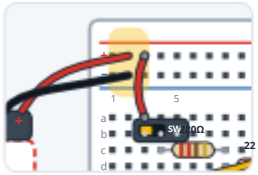
4. Jumper from row e, column 4 to row a, column 16. Ring 2 resistors: R3 (row b, 16–20) and R4 (row c, 20–24).



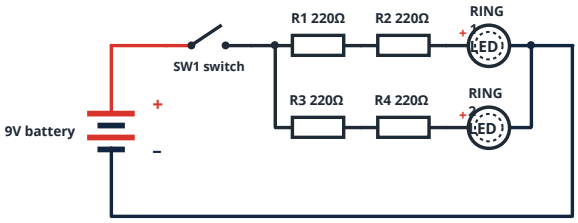
5. Clip crocodile wires to ring 2: the + one to a jumper in row d, column 24, the – one to a jumper in the – rail, column 26.



6. Connect the battery. Slide the switch ON. Both rings light up.



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

The switchfeeds two paths side by side, so both rings get the full battery voltage. Each path has its own resistors so each ring stays safe.

This is a parallel circuit. If one ring is unplugged, the other keeps glowing.

Parallel: paths side byside, each with fullpower.

Installation: a light display made for people to look at.

EXPERIMENT ZONE

PREDICT:Whatwill happento ring 2 if you unclip ring 1?

Test it. Which part of the circuit does each ring depend on?

ENGINEER CHALLENGE
Arrangetheringstomakeaface, an eye or a planet. Describe your display.

★ INVENTOR CHALLENGE
Add a button soyoucanturnon each ring separately. Draw your new plan first.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

My display looks like... It works because...

19

THE SOUND EXPLORER

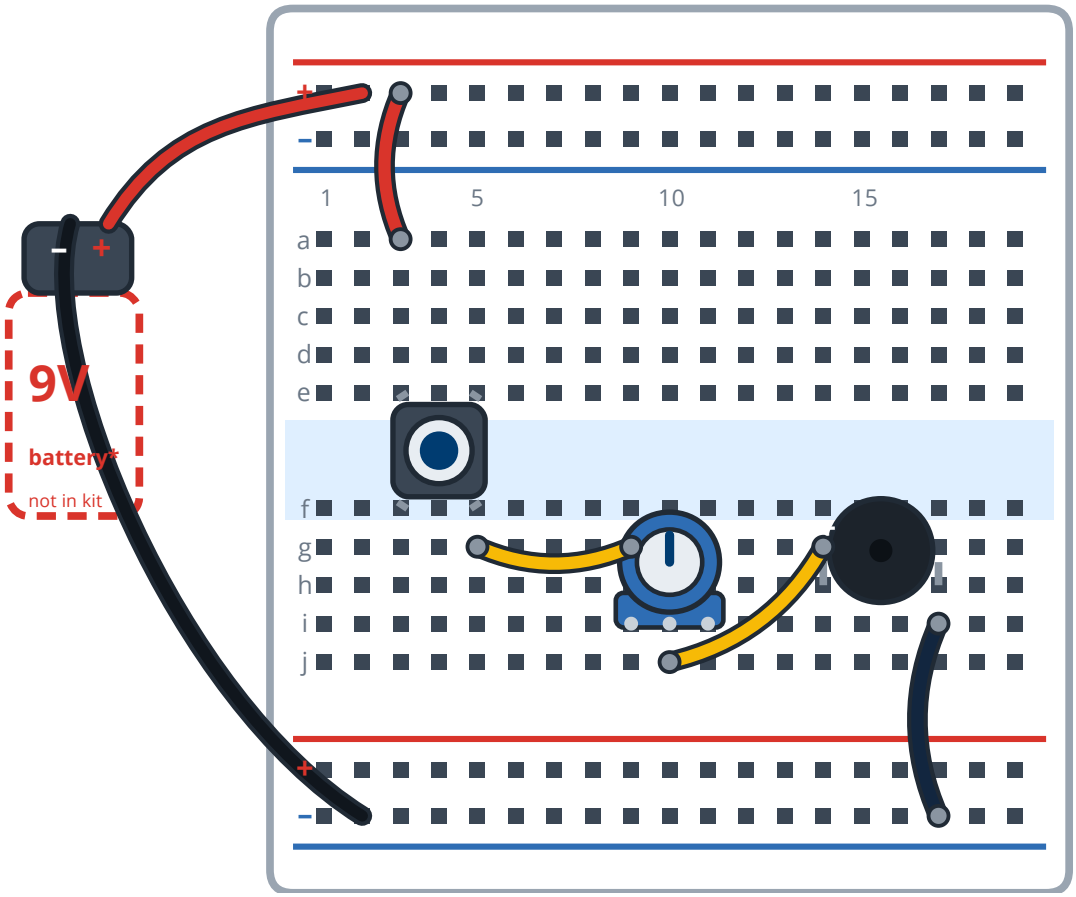
EXPLORER ENGINEER **INVENTOR**

STAMP **SOUND ENGINEER** earn it here

MISSION BRIEF

Amusician hearsa hum from the lab and wants to control how loud it is. Build a volume control for the buzzer and explore how electricity makes sound.

WHAT WILL YOU BUILD?



Close-up of the first columns of your breadboard. Yellow wires carry signals, red = +, dark blue = –.

YOUR TOOLKIT

- ×1 Push button
- ×1 Potentiometer
- ×1 Buzzer
- ×4 Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How changing the electricity changes the loudness of a buzzer.
- ★ How a buzzer needs a minimum electrical push to work.
- ★ How to measure sound without special equipment.

EXPLORER Follow the instructions and build.

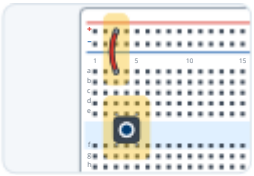
ENGINEER Modify thedesign and experiment.

INVENTOR Solve a problem independently.

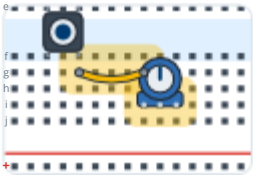
19 THE SOUND EXPLORER

BUILD IT

1. Press the button across the gap at columns 3 to 5. Jumper from the + rail to row a, column 3.



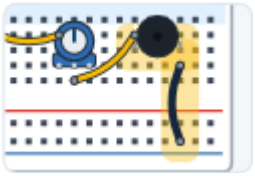
2. Jumper from row g, column 5 to row g, column 9. Push the potentiometer legs into row i, columns 9, 10 and 11.



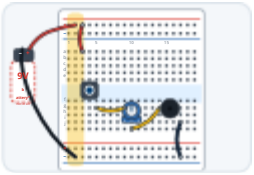
3. Jumper from row j, column 10 to row g, column 14.



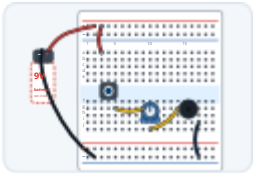
4. Push the buzzer legs into row h: the leg marked + in column 14, the other in column 17. Jumper from row i, column 17 to the bottom – rail.



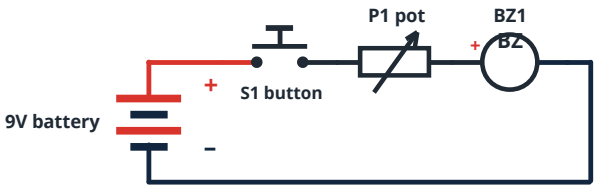
5. Connect the battery. Turn the knob to one end, then press and hold the button.



6. Slowly turn the knob. Listen. Where does the sound get quiet? Where does it stop?



CIRCUIT BLUEPRINT



positive (+) supply
ground / negative (–)
signal connection

Read it:
Follow the red line from + through each part to –. LED triangle points the way electricity flows.

WHAT'S HAPPENING?

Theknob changes how much electricity reaches the buzzer. More electricity makes a louder sound. Below a certain level the buzzer cannot work at all.

A buzzer is an electrical signal turned into vibration. Touch the buzzer lightly while it hums and you can feel it shake.

Volume: howloud a sound is.

Vibration: very fast shaking back and forth.

EXPERIMENT ZONE

PREDICT: Will thesoundget quieter smoothly, or switch off suddenly?

Turn the knob slowly. Describe what you hear at each position.

ENGINEER CHALLENGE
Put afingergentlyonthebuzzer while you turn the knob. What do you feel?

★ INVENTOR CHALLENGE
Design a volume controlforatoy. How could a child set it once and not change it?

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

Quietest setting that still buzzed was...

20

THE ULTIMATE INVENTOR CHALLENGE

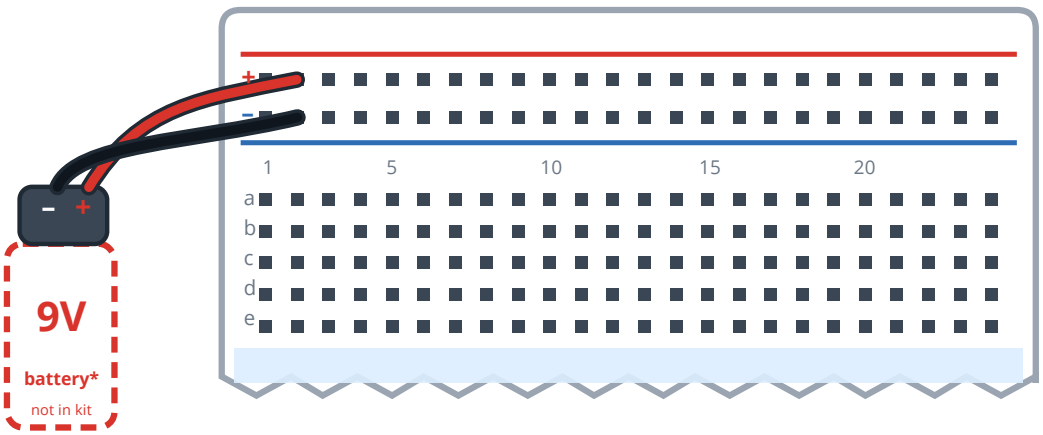
EXPLORER ENGINEER **INVENTOR**

STAMP
CREATIVE INVENTOR
earn it here

MISSION BRIEF

This is it Inventor.Noinstructions this time. Choose a real problem, design an invention that solves it, build it and test it.

WHAT WILL YOU BUILD?



NIGHT-LIGHT
Dimmer+LED.Who needs it and why?

SMART DOORBELL
Button + buzzer +light.How will visitors know it worked?

COOLING MACHINE
Fan +switch + lamp.Howwill you show it is running?

SECRET ALARM
Tapetripwire+buzzer. What will it guard?

Your breadboard is ready. Add your own parts, wires and a power connection to bring your invention to life.

YOUR TOOLKIT

- ×10 Resistors (220Ω & 10kΩ)
- ×8 LEDs (4 colours)
- ×1 RGB LED
- ×5 Push buttons
- ×1 Switch
- ×1 Potentiometer
- ×1 Buzzer
- ×1 DC motor
- ×1 Propeller
- ×2 LED rings
- 1 m Aluminium tape
- ×6 Crocodile wires
- strip Jumper wires
- ×1 Breadboard
- ×1 9V connector
- ×1 9V battery*

WHAT WILL YOU DISCOVER?

- ★ How to turn a problem into a plan.
- ★ How to choose an input and an output for an invention.
- ★ How to test, improve and explain your own design.

EXPLORER
Follow the instructions and build.

ENGINEER
Modify thedesign and experiment.

INVENTOR
Solve a problem independently.

20 THE ULTIMATE INVENTOR CHALLENGE

BUILD IT

1. PROBLEM: choose something that annoys you or someone you know.

1

2. IDEA: decide what your invention does. It needs an input (a button, switch or knob) and an output (a light, sound or motion).

2

3. PLAN: choose parts from your kit and draw your circuit in the blueprint box.

3

4. BUILD: build it one part at a time. Connect the battery last.

4

5. TEST: does it solve the problem? Write what works and what does not.

5

6. IMPROVE: change one thing and test again. Then show someone and explain it.

6

CIRCUIT BLUEPRINT

DRAW YOUR CIRCUIT HERE

Parts budget
Resistors: 10 in total (split to be confirmed). Buttons: 5. LEDs: 8. LED rings: 2. Everything else: 1 each. Draw each part with its symbol.

WHAT'S HAPPENING?

Every invention starts with a problem. Engineers draw a plan, build, test and improve many times before the final version works. Mistakes are normal. Each failed test shows you something new about how your design works.

Input: howyou controlaninvention.

Output: what the invention does.

EXPERIMENT ZONE

PREDICT beforeyoutest: what do you think will go wrong first?

Test your invention. Write what happened and what you changed.

⚙ ENGINEER CHALLENGE
Make yourinventioneasiertouse: add a label, a switch or a clear instruction.

★ INVENTOR CHALLENGE
Makeasecondinventionthat uses a different input and a different output, and compare them.

INVENTOR'S LOG · draw your design

My prediction was...

What actually happened...

Next time, I would...

My invention solves... It works by...

MY INVENTOR'S LOG

MY INVENTION'S NAME

DATE

WHAT PROBLEM DOES IT SOLVE?

COMPONENTS I USED

- [ ] Resistors
- [ ] Switch
- [ ] Propeller
- [ ] Jumper wires
- [ ] LEDs
- [ ] Potentiometer
- [ ] LED rings
- [ ] 9V connector
- [ ] RGB LED
- [ ] Buzzer
- [ ] Aluminium tape
- [ ] Push buttons
- [ ] Motor
- [ ] Crocodile wires

MY CIRCUIT SKETCH

HOW IT WORKS

WHAT I WOULD IMPROVE

MISSION TRACKER

Colour in each mission circle when you finish it, then stamp the matching badge.

CIRCUIT EXPLORER — MISSIONS 1–5

01	02	03	04	05
First Light	Traffic Light Controller	Rgb Colour Laboratory	Brightness Master	Button-Controlled Lamp

INTERACTIVE ENGINEER — MISSIONS 6–10

06	07	08	09	10
Two-Button Security Lock	Either-Or Switch	Intruder Alarm	Morse Code Communicator	Emergency Beacon

MOTION MAKER — MISSIONS 11–15

11	12	13	14	15
Mini Electric Fan	Reverse Engineer	Machine Start Button	Wind Power Lab	Control Centre

MASTER INVENTOR — MISSIONS 16–20

16	17	18	19	20
Conductivity Detective	Light And Sound Alarm	Dual Light Display	Sound Explorer	Ultimate Inventor Challenge

STAMP COLLECTION

CIRCUIT BUILDER	LIGHT MASTER	LOGIC DETECTIVE	SOUND ENGINEER	MACHINE MAKER	PROBLEM SOLVER	CREATIVE INVENTOR
M 1, 5	M 2, 3, 4, 10, 18	M 6, 7, 16	M 9, 17, 19	M 11, 12, 13, 15	M 8, 14	M 20

STEMverse

OFFICIAL STEMVERSE INVENTOR

This certifies that

INVENTOR'S NAME

has completed the STEMverse Makers Kit Explorers Edition

by building, testing and inventing across 20 missions.

MISSIONS COMPLETED

1 2 3 4 5 6 7 8 9 10

11 12 13 14 15 16 17 18 19 20

You tried. You tested. You fixed. You invented.

DATE

STEMverse